When Does Verification Help? Recovery-Path Regularization for Strategy-Aware Valuation

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Abstract

Recovering withheld information can increase a seller's predicted price when the valuation model is nonmonotone. We study this failure under natural missingness, charged failed queries and shared verification capacity. A recovery-path decomposition separates information recovery from its effect on concealment gains. We introduce recovery-path regularization, which penalizes increases in positive concealment gain along legal recovery edges while retaining public-record deployment inputs. A restricted information frontier states the limits of verification, rather than predicting learned-model performance. In a matched ten-seed study at 10% query capacity, the declared unit-weight endpoint reduces mean inflation by 50.41%, 26.58% and 23.81% on synthetic, JUCars and MUCars relative to same-gap zero-weight controls. All thirty source–seed comparisons have the same direction. Synthetic honest mean absolute error rises 0.82% and failed queries rise 1.10 percentage points, so lower inflation does not imply multiobjective dominance. Stronger training-response search also removes some earlier feedback damage, showing that the negative feedback result depends on implementation. The claim-bearing extensions complete a four-stage computational audit covering 1,206 distinct cells, including the 540-cell ten-seed study. Remaining response regret, retrospective cohorts and cross-machine numerical differences bound the findings. The supported practice is to diagnose recovery paths and evaluate incentive gains jointly with prediction and verification costs before deploying feedback or regularization.

Keywords: strategic missingness; secondhand valuation; budgeted verification; incentive auditing; factorial evaluation

1 Introduction

Secondhand valuation must use incomplete seller-provided descriptions while allowing for deliberate withholding. A vehicle with no recorded mileage and a vehicle whose mileage was suppressed can present the same public report. A platform may request additional evidence, but available verification capacity is finite and a naturally absent field may remain unavailable after a paid attempt. This study models the resulting interaction between an itemwise price predictor and a batch-coupled verification policy. The outcome is an advertised price prediction, rather than a transaction price or a transfer mechanism.

A query need not reduce the incentive created by an incomplete report. Consider an illustrative nonmonotone predictor whose quotes are 10 for a truthful record, 12 after two fields are withheld,

and 14 after only one is recovered. Successful partial recovery then raises the quote. These illustrative quote units are not empirical measurements. They expose a distinction between obtaining information and using the recovered record to reduce a seller's expected gain. Moreover, changing one report can displace another listing from a hard-capacity query set. Information value evaluated in isolation does not capture that competition.

Strategic withholding has established regression and classification treatments [9, 5]. Audits and partial-verification mechanism design likewise have established incentive guarantees under their respective action and outcome models [4, 6]. These results motivate verification, but they do not imply that an error-reduction acquisition score, a fixed library of query rules, or a finite-round feedback-training implementation will improve continuous valuation after failed-but-charged recovery. Ordinary active feature acquisition also targets predictive information rather than this strategic quotation objective [12].

We study two complementary questions. First, how does a uniform honest-precision requirement limit the strategic gain that a hard verification budget can remove when natural and strategic absence are observationally identical? Second, which implemented training and allocation choices actually account for changes in valuation outcomes?

A finite oracle enforces common decisions for identical public reports. A recovery-path decomposition separates the initial report, local recovery and capacity competition. A history-dependent cap provides a restricted unilateral-gain guarantee. A matched factorial experiment separates initial objective, acquisition score, policy feedback and allocation.

The central contribution is a recovery-path diagnosis linked to a targeted training intervention. Section 4 delimits guarantees under observationally identical missing reports; its restricted theorem is not a quantitative prediction for the learned four-field models. Section 5 decomposes the valuation effect of recovery, gives a conditional history-cap guarantee, and defines recovery-path regularization (RPL).

The empirical evidence separates initial objective, acquisition score, policy feedback and allocation. The original factorial identifies a weak-search feedback failure. Matched stronger-search interventions qualify that finding, while competitive ordinary-benefit and acquisition-conditioned controls prevent a universal claim for the original SAV candidate. The ten-seed RPL extension then tests the diagnosed recovery failure directly against same-gap zero-weight controls, including zero-query outcomes and path/gap removals. Its lower inflation coexists with predictive and verification costs. The claim-bearing extensions have completed a four-stage computational audit, with shared model primitives and conditional uncertainty explicitly disclosed. The resulting contribution concerns recovery-aware learning and a reusable comparison protocol, without establishing universal superiority or actual seller behavior.

2 Related work

2.1 Strategic withholding and predictive missingness

Zhang et al. studied strategic missingness in regression through support-margin imputation [9]. Krishnaswamy et al. characterized truthful classifiers and constructed predictors resistant to feature withholding [5]. These works already establish that natural absence and deliberate concealment can have the same report. We therefore do not claim that continuous prediction or ambiguous missingness defines a new problem. Our analytical question adds a uniform honest-quote error constraint and a hard query budget. Removing the incentive by changing predictions may violate that precision constraint, which is distinct from average mean absolute error.

2.2 Verification, randomization and information acquisition

Estornell et al. minimize incentives through budgeted audits, including threshold and top-k settings [4]. Zhang et al. study automated classification mechanisms with partial verification, including randomized decisions and combinatorial costs [6]. Marginal-probability representations are established in security-game optimization [3]. The finite oracle here specializes those tools to public report classes and continuous quote recovery, with failed queries charged against capacity. Its complete finite coverage is a certificate within the declared world universe, not a guarantee for unseen types. Active feature acquisition addresses budgeted information selection [12]; its predictive objective differs from the signed unilateral quotation difference audited here.

Recent acquisition methods include amortized greedy selection (GDFS) [17], discriminative information estimation (DIME) [18], stochastic latent encodings (SEFA) [19], and explanation-ranked transformer policies [20]. Alongside ACO [12], they broaden acquisition coverage beyond simple risk and predictive benefit. Their predictive objectives, sequential budgets and input assumptions do not imply control of strategic quotation gain. Our executed adapters and their limitations are specified in Appendix H.

2.3 Strategic feedback and contextual policy classes

Performative prediction analyzes retraining when deployment changes the data [10]; strategic empirical risk minimization differentiates through modeled user responses [11]. Contextual games use public side information to condition decisions [7]. Joint score and audit design is also explicit in recent audit-risk work [8]. We do not transfer their convergence conditions to our nonconvex, finite-round training loop or introduce randomization as a new technique. Instead, we test an epoch-refreshed implementation under matched fitting budgets and separately intervene on within-epoch score refresh and training response-search strength. Our constructed policy-library separation also includes a stronger small context-aware comparator that closes the gap. This prevents a restricted library from being mistaken for evidence that dynamic weights or an LP solver are intrinsically necessary.

3 Problem formulation

3.1 Reports, queries and seller utility

A batch contains $n = 100$ listings. Listing i has a latent recorded feature vector x_i, including four optional action fields: mileage, transmission, fuel type and body condition. Let M_i be its naturally absent fields. A seller chooses a concealment set s_i containing at most two available action fields. Its public report $r_i(s_i)$ replaces fields in $M_i \cup s_i$ with missing values. Available values cannot be fabricated, changed or recovered from public identifiers. A missingness indicator does not disclose membership in M_i.

The platform observes the whole report batch $r = (r_1, \ldots, r_n)$, locks the reports and chooses a possibly randomized allocation. It may query at most one publicly missing field per selected listing, with at most $B = \lfloor bn \rfloor$ queries on every realized draw. A query of a concealed recorded field recovers that value; a query of a naturally absent field fails. Both spend one unit. A fixed itemwise model produces log quotes before and after the selected query. Scores and allocations at deployment may use public reports and model outputs, but not the latent recoverability indicator or the advertised target.

Seller utility is the expected log quote minus concealment cost. Main empirical costs are zero; all randomization occurs after reports are locked. Evaluation uses finite response search and explicitly

checks every legal single-seller report against fixed peer reports, reallocating the batch each time. A returned profile with small remaining regret is a local deviation diagnostic, not a certificate of a globally selected market equilibrium. Predictions do not set a transaction price, and buyer preferences and bilateral trade are outside the model.

3.2 Predictors, public scores and comparator identity

Suppression-aware valuation (SAV) combines an augmented price predictor, a suppression acquisition head and simulated query-feedback training. The original static-robust comparator changes the prediction training objective without that feedback procedure. Counterfactual deviation-constrained mixture verification (CDM) instead fixes the price model and calibrates probabilities over complete budget-feasible query rules. Its calibration objective constrains stored own-report/peer-profile rows, not all unseen types. The original library contains twelve rules and five calibration peer-profile families; expanding the finite library does not itself establish out-of-sample incentive compatibility. These original candidates provide the context for the matched experiment, whose newly trained initial-objective factor must not be renamed SAV or static robust.

For training targets, let q_s be a cross-fitted quote under report state s, $q_{s\setminus f}$ the quote after recovering field f, q^H the honest quote, and y the log advertised price. The three target quantities are

$$v_{sf}^{\mathrm{sup}} = [q_s - q^H]_+ - [q_{s\setminus f} - q^H]_+, \quad v_{sf}^{\mathrm{ben}} = (q_s - y)^2 - (q_{s\setminus f} - y)^2, \quad v_{sf}^{\mathrm{risk}} = (q_s - y)^2.$$

A common multioutput ExtraTrees head learns these quantities from encoded public inputs [14]. Honest records and targets construct training labels only; deployment does not directly substitute them into the scores. Each policy chooses its highest-scoring publicly eligible field and queries only positive scores. Top-B ranks these per-listing scores, using the declared stable tie rule. The score-minimax control instead balances positive residual scores through query marginals. It solves a score surrogate, not the true unilateral incentive problem.

3.3 Evaluation quantities

For one returned report, let q^H be its model's honest log quote, y the log-transformed advertised price, q_0 the no-query quote, q_1 the quote conditional on the selected query, and p its marginal query probability. Here $y = \log(1 + \text{price})$ and $\mu = (1 - p)q_0 + pq_1$. The incentive coordinate is $[\mu - q^H]_+$; it compares the model with itself, rather than with the advertised target. The separate realized-positive quantity is $(1-p)[q_0-q^H]_+ + p[q_1-q^H]_+$. Honest price error is $|\exp(q^H) - \exp(y)|$, and attacked price error is

$$(1 - p)|\exp(q_0) - \exp(y)| + p|\exp(q_1) - \exp(y)|.$$

The subtraction of one cancels inside each absolute difference. Taking an absolute error after averaging the quotes would define a different metric. Failed-query probability is p times the indicator that the selected field cannot be recovered; such an attempt still consumes capacity. Signed log bias is $\mu - y$, while expected squared log error averages the two squared endpoint errors before aggregation.

Inflation alone admits a degenerate solution. A constant-quote model has zero inflation for every report, irrespective of its honest or attacked price error. Thus zero inflation is not evidence of useful valuation. We treat inflation, attacked MAE and failed-query probability as co-primary

descriptive outcomes, with honest MAE as the precision coordinate. The joint outcome vector, continuous paired effects and Pareto trade-offs take priority over any single ranking. This reporting choice is retrospective; it is not a preregistered change in the original experiment.

For the unilateral audit, hold peers' returned reports fixed and enumerate all feasible reports of one seller, reallocating the whole batch for each. Writing $u(r)$ for that seller's expected log quote minus concealment cost, the deviation envelope is $[\max_r u(r) - u(r_{\text{honest}})]_+$; remaining regret instead subtracts the utility of the returned report. Thus a small regret does not establish a small gain over honesty. Main empirical concealment costs are zero. All quantities are aggregated over the retained listings; nonconverged responses remain in the analysis. The finite oracle's constructed quote units must not be interpreted as empirical log-price effects.

4 Information limits and finite reference mechanisms

4.1 Public-information precision frontier

Takeaway. A finite complete oracle can distinguish a policy-library restriction from the best value achievable under the same public information.

Let o be the complete observable batch signature and $A_B(o)$ its feasible complete nonadaptive allocations, each spending at most B queries. Allocate probability z_{oa} to action a, sharing the same variables across latent worlds with the same signature. For each legal deviation d, let $G_d(z)$ be the expected deviated quote minus the truthful quote, with the entire batch reallocated on both sides. The finite oracle solves

$$\min_{z,t} t \quad \text{s.t.} \quad G_d(z) \le t \ (\forall d), \qquad \sum_{a \in A_B(o)} z_{oa} = 1, \quad z_{oa} \ge 0, \quad t \ge 0.$$

This is exact only for the declared complete finite world/report/action universe. Its numerical primal/dual certificate uses tolerance 10^{-8}. It is not a guarantee over unseen types or general adaptive query trees.

For the frontier, we optimize both verification and the itemwise quote function subject to the precision constraint; a fixed empirical predictor need not attain the bound. Consider a separate alias family with an all-natural-missing world and worlds in which only listing i has a recoverable low-value field, lowering its target by $\Delta_i > 0$. Withholding it produces the same public batch. Require verification-neutral quotes (same final record, same quote), and uniform absolute error at most η on all specified honest reports. Set $a_i = (\Delta_i - 2\eta)_+$. The attainable worst positive expected gain is

$$T^*(\eta, B) = \min_{0 \le p_i \le 1,\ \sum_i p_i \le B} \max_i a_i(1 - p_i). \tag{1}$$

For the nonzero interior case, $\sum_{i:a_i>0}(1 - T^*/a_i)_+ = B$.

Proof and boundaries. Couple queries by their private random seed on the all-failure transcript. Let p_i be the probability of querying i. Hard capacity gives $\sum_i p_i \le B$. By verification neutrality and the fixed itemwise quotation rule, unsuccessful queries do not change the quote. Until i is queried, its hidden low-value world has the same transcript; without recovery, the honest-precision constraint leaves a quote gap at least $\Delta_i - 2\eta$. This proves the lower bound, including adaptive policies in this single-recoverable family. Choose quote shifts $e_i = \min(\eta, \Delta_i/2)$ toward one another and marginals $p_i = (1 - T^*/a_i)_+$. Systematic interval sampling realizes these marginals with at most B queries per draw and attains the bound. This does not follow from an average MAE bound. It does not imply zero adaptivity benefit for failed-query counts or multi-field worlds.

What the frontier excludes. Let $k_\eta = |\{i : \Delta_i > 2\eta\}|$. Equation (1) implies $T^*(\eta, B) = 0$ if and only if $B \geq k_\eta$: each strictly positive residual gap needs unit query probability. In particular, at $B = 0$ zero gain requires $\eta \geq \frac{1}{2}\max_i \Delta_i$. For n identical gaps Δ, the frontier simplifies to

$$T^*(\eta, B) = (\Delta - 2\eta)_+ \left(1 - \min\{B/n, 1\}\right).$$

The lower bound follows because some marginal is at most B/n; equal marginals attain it. Thus honest precision, query capacity and strategic gain cannot be specified independently in this alias family.

This is compatible with existing truthful predictors. Natural-versus-strategic missingness ambiguity is explicit in [5], whose Theorems 1 and 2 give minimum-cut optimality and a MAX-ensemble characterization for binary classification. A predictor can eliminate withholding gains by changing its predictions, while violating the uniform error requirement imposed here. Our lower bound restricts that escape route; it does not contradict truthful classification or establish impossibility under average MAE alone. History-dependent quote rules satisfy this frontier only when they also meet its verification-neutrality and uniform-precision assumptions on the specified family. An arbitrary history cap cannot be placed inside that mechanism class without checking these conditions.

A two-mode constructed example at $B = 1, \eta = 0$ yields 72/17 for the public oracle, 72/13 for the best global mixture of eight fixture rules, and 8 for risk ordering. The eight rules are the six priority permutations of three items, risk ordering by residual leverage $(\Delta_i - 2\eta)_+$, and no query; this fixture library differs from the twelve-rule empirical CDM library in Section 3.2. These are artificial linear quote units, not empirical log-price effects. All 32 budget/precision cases passed a separate auditor.

What is tested in the learned-model experiment. Equation (1) applies to the single-recoverable alias family, verification-neutral quotes and a uniform error constraint. The empirical model has four fields, can hide two and is evaluated by average MAE, so these assumptions do not establish its empirical optimum or predict a feedback sign. We use the frontier as a scope boundary. The recovery analysis instead gives a finite diagnostic on the actual model class: enumerate legal partial recoveries and test whether a query raises the quote above both its pre-query and honest anchors. Section 7.1 performs that check.

4.2 Compact implementability and contextual comparators

Takeaway. Item-local utility permits an exact compact oracle; enlarging a globally mixed priority library need not remove its contextual limitation.

Let y_{oif} be the probability of querying field f of item i in public report class o. Impose

$$y_{oif} \geq 0, \qquad \sum_f y_{oif} \leq 1, \qquad \sum_{i,f} y_{oif} \leq B,$$

and zero probability on publicly ineligible fields. Every declared unilateral deviation has the affine form $G_d(y) = b_d + \sum_{o,i,f} c_{doif} y_{oif}$ when its realized utility depends only on its own queried field or no query. Coefficients include both truthful and deviated report classes. Compatible latent worlds must share the same y variables.

Proposition 1 (Compact finite oracle)**.** *For item-local utilities, nonadaptive queries, unit query costs and at most one queried field per item, minimizing* $\max\{0, \max_d G_d(y)\}$ *over the above constraints has the same value as the complete-allocation lottery oracle. Every feasible marginal solution is implementable under a hard budget on every draw.*

Proof. Every complete-allocation lottery induces feasible y. Conversely, fix a public class and set $p_i = \sum_f y_{if}$. Arrange consecutive intervals of lengths p_i. A private uniform $U \in [0,1)$ selects half-open intervals containing some $U + k$, for integer k. Each interval has length at most one and the total length is at most B, so at most B items are selected; item i has selection probability p_i. For $p_i > 0$, conditional on selection, draw its field with probability y_{if}/p_i. Thus every field has exactly the prescribed marginal and all affine expected deviation payoffs are preserved. □

With O public classes, n items and F fields, this representation uses at most $OnF + 1$ variables rather than one variable per complete allocation. It does not remove counterfactual-world or report enumeration, preserve arbitrary joint event probabilities, or cover adaptive queries and nonunit costs. The history cap is item-local and therefore admits the same reduction. Marginal formulations and their implementability limitations are established in security games [3]; the proposition gives the specific implementation argument needed for this valuation audit, not a claim to invent marginal optimization.

Independent checks match compact and complete-enumeration values on all 32 original market fixtures and 24 multi-field affine fixtures. One larger algebraic fixture has 100 items, 4 fields, 8 public classes, 96 payoff rows and budget 10. Its compact LP has 3,201 variables, versus 18,662,296,388,649,543,921 complete allocations per class. All 57 numerical primal/dual certificates pass. The repaired sampler has 3,200 exactly checked rational field marginals across 111 interval pieces; 16,000 numerical draws respect hard capacity. The large full-enumeration solver was not run, so this is a representation comparison, not a measured speedup.

The contextual-comparator result is a limitation of rule information, not an LP-versus-heuristic theorem. In the constructed family, a mixture over all context-insensitive priority rules has an unbounded capacity-related gap against context-conditioned allocation. A fixed mixture of only $B + 1$ context-aware rules closes that gap exactly. Appendix A gives the construction, proof and exact witnesses. Both comparators must be retained when interpreting the finite oracle.

5 Recovery paths and a conditional history cap

5.1 Signed mechanism decomposition

Takeaway. Freezing truthful queries cannot separate recovery from capacity in this replay model; use a capacity-relaxed public query instead.

For each fixed peer profile and evaluated mixture, retain its saved maximizing own report s^*. Let q^H be the truthful quote, q_h the unqueried reported quote, q_L the quote after the same rule's highest-scoring eligible query with capacity relaxed, and q_A the hard-budget quote. Ineligible or nonpositive scores still cause no query in the relaxed reference. Then

$$E = q_h - q^H, \qquad R = \mathbb{E}[q_L] - q_h,$$
$$C = \mathbb{E}[q_A] - \mathbb{E}[q_L], \qquad G = E + R + C.$$

All components are signed and evaluated at the same s^*. Mixture expectation precedes report maximization. The relaxed query is a diagnostic, not a feasible deployment policy. The original truthful-query anchor would give zero recovery: that query targets a naturally absent field, which remains absent when another field is withheld. Its remaining term would conflate recovery and capacity.

Across all nine source/seed pairs and both price models at 10% capacity, 360,000 component rows were reconstructed separately by complete stable sorting of each hypothetical batch. The

audit checked 11,343,240 legal rule–world allocations and rejected four output mutations. Eighty-one saved winners differed from freshly computed argmax labels only through floating ties; the largest payoff gap was 6.94×10^{-18}. Original winners were retained. The initial argmax-identity failure is preserved in a separate run.

In the illustrative 10/12/14 example, $E = 2$ and $R = 2$. If the item receives its partial query with probability p, then $C = -2(1-p)$ and $G = 2+2p$. The cap leaves the reported quote at 12 on both branches, reducing this gain to 2 rather than making honesty optimal. Positive R can occur: hiding two fields and recovering one can move a nonmonotone estimator to an intermediate record with a higher quote. This motivates the following developmental mechanism experiment. It is not evidence that all original SAV advantages or failures have been causally explained.

5.2 History cap and fixed-peer guarantee

Takeaway. Removing quote increases along the realized recovery path gives a conditional unilateral-gain guarantee without changing truthful quotes.

For every realized allocation, use

$$q_{\text{cap}}(r, a) = \min\{q(r, \varnothing), q(r, a)\}.$$

Apply this operation before averaging over a policy mixture. The underlying public scores, query budget, model and mixture weights are fixed.

Proposition 2. *If truthful queries cannot alter the fixed itemwise quote, then for any fixed peer profile and any fixed policy mixture the cap preserves the truthful quote and weakly reduces the worst expected unilateral gain over legal own reports.*

Proof. Pointwise $q_{\text{cap}}(s, a) \le q(s, a)$, whereas $q_{\text{cap}}(0, a) = q(0, a) = q^H$ for every realized action. Subtracting the same truthful quote, taking expectations and then maximizing over legal s preserves the inequality. The null deviation yields zero. □

The mechanism deliberately uses verification history. It lies outside the verification-neutral class in Equation (1), so it cannot be used to claim a violation of that bound. Equal final records can have different prices under the cap. The guarantee concerns fixed peers and scores; a new seller-response profile can change allocations and outcomes. Lower quotes need not have lower absolute pricing error.

The full comparison includes both estimators and all six capacities: 108 cases, 5,671,620 legal report rows and 2,160,000 entry–method comparisons. A separate auditor reconstructs absolute quotes rather than reusing the signed-gain cap calculation. All truthful quotes are unchanged and all fixed-profile envelopes weakly decrease within tolerance 10^{-10}. Risk has zero improvement in the primary fixed-profile table in Appendix B and is retained as a strong reference. These are additional conditional diagnostics, not new ten-seed training experiments.

5.3 History dependence and a final-record alternative

The cap can assign different quotes to identical final records because it retains the pre-query quote. Consequently, final-record consistency cannot be claimed. A platform would need to explain the history rule, allow correction and review, and evaluate whether verification selection creates systematic disparities. We have not measured protected-group effects or established a deployment fairness guarantee.

A final-record rule can instead be defined on a declared completion domain $\mathcal{Z}$. Let $\mathcal{C}(r)$ contain its records extending the observed values of r, and define

$$q_{\downarrow}(r) = \min_{z \in \mathcal{C}(r)} q(z).$$

If a truthful record belongs to the domain, every concealed or partially recovered report extends to that record, and the completion sets are nested. Hence $\mathcal{C}(r^H) \subseteq \mathcal{C}(r)$ and $q_{\downarrow}(r) \leq q_{\downarrow}(r^H)$. Nonnegative concealment costs therefore prevent a gain over this rule's own truthful quote, regardless of which admissible partial query is allocated. The rule uses only the final record and its fixed domain. This is an order-preserving envelope, not a new general imputation technique; truthful withholding predictors are established prior work [5, 9].

The alternative can lower honest quotes and worsen honest MAE, unlike the history cap's truthful-quote preservation. Its guarantee needs completion-domain coverage and nonempty nested sets; unsupported records require abstention or an explicitly uncertified fallback. A three-seed categorical-domain pilot quantifies one limited honest-error cost: MUCars shared-support MAE increases by 10.49–23.87 currency units, concentrated among 28 naturally incomplete records (Appendix G). Mileage is observed and fixed, so this is not a tested four-field replacement (Appendix G).

5.4 Recovery-path regularization of the public predictor

Takeaway. A training penalty can target the diagnosed recovery failure directly, but a finite penalty is not an incentive guarantee.

Let $q_\theta(s)$ denote a log quotation and $q_\theta(H_i)$ its honest-report anchor for training record i. For each legal hidden-field recovery edge $(s, a) \in \mathcal{E}_i$, define

$$g_{i,\theta}(s) = [q_\theta(s) - \mathrm{sg}(q_\theta(H_i))]_+, \tag{2}$$

$$L_{\mathrm{path}}(\theta) = \frac{1}{\sum_i |\mathcal{E}_i|} \sum_i \sum_{(s,a) \in \mathcal{E}_i} [g_{i,\theta}(T(s,a)) - g_{i,\theta}(s)]_+. \tag{3}$$

Here sg stops the anchor gradient within an optimizer step. The objective adds $\lambda L_{\mathrm{path}}$ to the matched no-feedback loss; $\mathcal{E}_i$ contains recoveries of actually concealed fields, excluding naturally unavailable values. The ordinary gap term penalizes a quotation relative to honesty; this term additionally penalizes increases in positive gain along intermediate recoveries. Honest values and the anchor are training supervision, not deployment inputs. Deployment uses the same public-record predictor, without a quotation-history input.

Proposition 3 (Finite recovery-graph implication)**.** *On a fixed finite collection of record graphs, if $L_{\mathrm{path}} = 0$, positive gain cannot increase along any path composed of their legal recovery edges.*

Proof. Every summand in (3) is nonnegative. A zero finite sum makes each edge increment nonpositive; chaining these inequalities proves the claim. □

This implication allows positive initial concealment gain. It provides no guarantee on unseen records, finite-penalty fits, allocation optimality or honest error. Shared parameters can move the anchor between steps despite its stopped gradient. Recovery-path regularization (RPL) is therefore a mechanism-targeted candidate, not a new general monotonicity principle or an established replacement for existing strategic-missingness methods.

6 Experimental design and corroborating checks

6.1 Cohorts and source boundaries

Takeaway. The evidence concerns advertised-price prediction on fixed retrospective cohorts and explicitly modeled seller responses.

JUCars-2024 version 2 contains Jordanian listings from 2024 and was deposited in 2026 [1]; MUCars-2024 version 2 contains Moroccan listings from 2024 [2]. Both sources are distributed under CC BY 4.0. Recorded prices are advertised prices, not completed transaction prices. We evaluate native currencies separately. The synthetic market follows the saved generator rather than an empirically identified transaction model. Before the present caps, the processed training/evaluation splits contain 4,774/1,374 JUCars records and 39,244/10,672 MUCars records. Identifier hashing caps training at 5,000 and evaluation at 1,000 records. Synthetic generation uses fixed seeds 73,101 and 73,102 for 6,000/1,500 records before the same caps. Fitting seeds vary training randomness, not the source cohorts or generator. The hash cap is deterministic and label-independent; all ten fitting seeds share exactly the same evaluation records and batch membership. Seed variation reflects fitting randomness only. The validation sources were inspected during development and are retrospective, rather than fresh confirmatory tests. The capped cohorts contain ten batches of 100; these small cluster counts restrict uncertainty claims.

Original advertisement identifiers are unavailable; inherited non-price fingerprints provide duplicate control without certifying independent vehicles. Targets, identifiers, source indices and declared direct reveal aliases are excluded from predictor inputs. Correlated covariates remain, so concealment need not remove all information about a field. Naturally absent fields cannot be concealed by a seller or successfully recovered by a query.

6.2 Matched training and evaluation

Takeaway. The full controlled experiment and four-stage audit are complete; controlled contrasts do not establish a general method advantage.

The new design crosses two initial training objectives (mask-augmented and robust), three acquisition scores (suppression, benefit and risk), two policy feedback settings, and two allocators (top-B and score-minimax). The initial objective factor is a new controlled intervention, not a relabeling of the original SAV and static-robust checkpoints. Each cell uses the same backbone architecture, six initial residual epochs and six subsequent epochs in two rounds, with matched record orders and head-refresh schedules. No-policy-feedback fits are shared across their six acquisition/allocation aliases, giving fourteen unique fits for twenty-four cells. The implemented sequence is:

1. For each of three outer folds, exclude its records from the complete training loop. Split the remaining records into estimator and head-guidance sets, using a guidance fraction of 0.3. Fit the augmented backbone and each six-epoch initial residual objective.

2. Clone each initial model into one no-policy-feedback fit and six score/allocator-specific feedback fits. Run two three-epoch robust-update rounds. Refit the guidance head after each round, retaining identical record orders across the matched cells.

3. At the start of each feedback epoch, compute public acquisition scores and hold them fixed during that epoch. Within each minibatch, use the current detached quotes to search seller responses, with two sweeps and one start at 10% capacity. Reallocate the batch for every candidate report. Update only the residual model using the resulting frozen response and

probabilities. The no-policy-feedback control instead searches valid reports with no queries; it still receives the same six robust-update epochs.

4. Predict every excluded fold with its corresponding trained model. Fit the deployment model through the same training sequence, then refit its 160-tree deployment heads using the assembled outer-fold quote targets and the deployment model's encoded inputs. These heads are used for evaluation; the deployment residual model is not updated after this final head fit.

5. Evaluate each fixed deployment model at capacities 0, 1, 5, 10, 20 and 30%, using four response starts and at most twenty sweeps. Compute expected quotes and errors analytically from the query marginals; retain every final profile and its independently checked unilateral regret.

This protocol does not refit a separate model for each evaluation capacity. Effects away from 10% therefore include transfer from the fixed training capacity. Evaluation uses stronger response search than training, but finite search still does not certify a global best-response equilibrium.

For the score-minimax control, an item's highest positive publicly eligible score is g_i. We minimize $\max_i g_i(1-p_i)$ subject to $0 \le p_i \le 1$ and $\sum_i p_i \le B$, then realize the marginals with a private systematic draw. This is a score surrogate and an established resource-allocation construction; it is not a guarantee about report-dependent unilateral incentives. Every candidate seller report therefore triggers a fresh batch allocation in the response calculation.

For a frozen within-step query marginal p, training uses the expected post-query squared error,

$$(1-p)(q_0-y)^2 + p(q_1-y)^2 = ((1-p)q_0 + pq_1 - y)^2 + p(1-p)(q_1-q_0)^2.$$

Taking squared error after averaging quotes would omit the variance term. The gain penalty instead uses the positive part of expected gain, matching the expected-utility objective. Gradients do not pass through the discrete report search, field selection or allocation probabilities; this remains a frozen-policy training surrogate.

Writing L_H for truthful squared error, L_V for mean squared error over valid report states and L_A for the squared anchor to the backbone's honest quote, the mask objective is $L_H + 0.25L_V + 0.25L_A$. The robust objective adds $0.5L_{\text{attack}} + 0.5L_{\text{gain}}$, where $L_{\text{gain}} = \mathbb{E}[(\mu - q^H - 0.02)_+]$ and L_{attack} averages the endpoint squared errors before expectation. The honest anchor in the gain penalty is detached. AdamW uses weight decay 10^{-4} and gradient clipping at norm 5. The residual has two width-64 ReLU hidden layers and a scalar output. The subsequent robust update combines truthful squared error, a quarter-weighted mean squared error over valid report states, a quarter-weighted anchor to the backbone's truthful quote, and half-weighted expected attacked error and gain penalty. The declared weights and twelve-epoch total are common across matched cells. The distinction between expected squared error and squared expected quote is necessary for correct stochastic replay training; the variance identity itself is not a new mathematical result.

For attribution, at fixed source, initial objective, acquisition score and capacity, let $L_{f,a}$ be an outcome under feedback setting f and allocator a. We report the interaction

$$D = (L_{\text{none,minimax}} - L_{\text{policy,minimax}}) - (L_{\text{none,top}B} - L_{\text{policy,top}B}).$$

For loss-like outcomes, positive D means feedback has a larger incremental benefit with minimax. Both constituent effects must also be shown: positive D alone does not establish that either feedback policy is beneficial. The fixed analysis contract contains 58 conditional factor contrasts per source/capacity, including six feedback–allocation interactions. The primary capacity therefore has

174 exploratory contrasts, all retained rather than screened for favorable directions. No individual unadjusted interval is treated as a confirmatory finding across this family.

The full experiment completed thirty source–seed pairs and 4,320 cells. The training audit reconstructed hash-ranked partitions, matched minibatch orders and 2,068,360 held-out prediction rows. It refitted all 420 deployment heads, comparing tree structure and leaf values, and rejected altered partitions, permutations and predictions. This audit verifies fit provenance and head targets, rather than replaying every optimizer step. The outcome audit checked 4,320,000 recorded outcomes and 45,372,960 legal unilateral deviations, including reallocation after each changed report. A separate statistical checker reconstructed 9,216 metric–contrast rows, validated contrast semantics and intervals, and rejected altered interaction signs and interval endpoints. The full four-stage completion receipt was issued on 13 October 2026.

Audit dependency and independent state coverage. The historical outcome audit is a corroborating re-derivation with shared primitives. It imports the separate replay checker and shares the production state representation, data loader, prediction and head-scoring APIs. It does not independently reimplement those APIs or every training-library operation. Its rejection of changed outputs therefore does not certify absence of a shared state-transition bug.

A separate bit-mask oracle now checks state membership, validity, materialized values, public missingness and query success/failure without using production state or query logic in the oracle. Production StateSpace is only the system under test. All 16 natural masks and maximum-hidden values 0 through 4 cover 1,188 fixture field transitions. Direct CSV checks on all 1,374 JUCars and 10,672 MUCars validation records cover another 521,164 legal field transitions. Four injected enumeration/transition/materialization faults are rejected. Model prediction, scoring, dataset construction and optimizer libraries remain outside this independent oracle's scope.

Table 1 records the declared settings.

Completion study and ten-seed extension. The three-seed development study completed 1,719 cells, including reused references, across component removals, parameter scans, acquisition adapters, search interventions, availability stress and the original RPL pilot. Appendix H retains the complete inventory. The subsequent RPL extension kept the six configurations and three budgets unchanged, reused the original 162 cells, and added seven fitting seeds per source. Its 540 cells cover thirty source–seed pairs. This increases fitting replication, not the number of independent training populations or evaluation cohorts. Parameter and availability scans retain their original three seeds.

Completed computational audit and its limits. The original 4,320-cell chain remains unchanged. A new fit/outcome/analysis/ statistics-check chain covers the 828 claim-bearing original extension cells and 378 additional RPL cells, hence 1,206 distinct cells. The audited stages are the RPL pilot (162), search interventions (90), SEFA/GDFS/DIME/explanation adapters (432), ACO including neighbor sensitivities (144), and the ten-seed RPL extension (540, overlapping the pilot by 162). Each stage reconstructs fitting roles and provenance, re-derives all recorded batch outcomes and legal unilateral deviations, aggregates comparisons, and separately reconstructs the saved statistics. Completion does not independently reimplement shared prediction libraries or provide external scientific confirmation.

The new analyses retain 4,000 paired crossed seed–batch resamples. Marginal percentile intervals and exploratory studentized maximum-deviation bands condition on the same fixed populations; the RPL family contains 216 metric contrasts. Neither the ten-seed nor the three-seed

Setting	Value
CatBoost backbone boosting rounds [13]	250
Residual hidden width / learning rate	64 / 0.001
Initial epochs / robust-update epochs	6 / 6
Outer folds / guidance fraction	3 / 0.3
Acquisition-head trees / maximum depth / minimum leaf	160 / 12 / 12
Training budget / response starts / response sweeps	10% / 1 / 2
Evaluation response starts / maximum sweeps	4 / 20
Batch size / maximum concealed available fields	100 / 2
Gain-penalty tolerance / main concealment cost	0.02 / 0
Original factorial seeds / bootstrap repetitions	10 / 1,000
RPL path weight λ / gap weight β	$0, 0.1, 0.5, 1$ / 0.5
RPL gap-removal path weights / gap	$0, 0.5$ / 0
RPL evaluation query capacities	$0, 0.1, 0.3$
Completion / RPL-extension fitting seeds	3 / 10
Core-extension audit resamples	4,000
Recent acquisition families / frozen price backends	5 / 2
Recent-method prediction bins	4
ACO neighbor default / fixed sensitivity values	25 / 10, 25, 50
Additional natural-missingness fractions	$0, 0.1, 0.3$

Table 1: Declared settings across experiments. The ten fitting seeds are 13, 42, 2026, 101, 211, 307, 401, 503, 601 and 701. The backbone and ExtraTrees implementations follow established methods; they are not new algorithmic contributions. All capacities share the fixed training budget. Three-seed completion experiments use the first three seeds; RPL extends to all ten. The additional settings distinguish original-factorial inference from the new audit families.

Table 2: Computational audit scope. Counts overlap where explicitly stated; human scientific approval and exact cross-machine reproduction are separate from these checks. Historical failure receipts are preserved.

Evidence	Completed chain	Boundary
Original 4,320 cells	Fit, outcome, analysis, statistics check	Original implementation; shared primitives disclosed
Claim-bearing extensions: 1,206 distinct cells	Same four computational phases; includes all 540 ten-seed RPL cells	Conditional exploratory inference; not an independent full-stack implementation
Remaining completion-study support	Hash bindings, utility checks and recorded summaries	No whole-study four-stage upgrade of all 1,719 cells

families have a finite-sample family-wise coverage certificate. The remaining parameter, component and availability cells are not silently upgraded to the new chain. All failed and nonconverged outcomes remain in their respective evidence sets.

7 Results

7.1 Recovery failures in the learned four-field class

The recovery-path prediction is observable in the saved models, rather than only in the illustrative 10/12/14 fixture. We enumerate all legal successful recovery edges in the 14 unique fits per source/seed and the original 1,000 evaluation records. An adverse partial recovery requires two hidden fields, a post-query quote above the pre-query quote and both at or above the model's honest anchor, using tolerance 10^{-10}. Table 3 reports these finite table properties separately from returned response rows.

Source	Successful edges	Adverse partial edges	Adverse queried rows
Synthetic	2,098,600	82,939	82
JUCars	2,044,000	67,523	1,733
MUCars	2,212,560	61,404	2,829

Table 3: Recovery-path diagnostic on saved primary-capacity models. Edges use 14 unique fits per source/seed; the last column counts returned reports from all 24 cells with positive query probability and gain-increasing partial recovery. For randomized allocation these are eligible rows, not realized query counts. Each source reuses the same 1,000 records over ten fitting seeds. Counts are dependent finite diagnostics, not independent samples or deployment prevalence.

A MUCars witness has honest, two-hidden and partially recovered log quotes 11.484823, 11.517865 and 11.546430, respectively. It hides transmission and fuel type, then recovers transmission. Thus its report component is about 0.033042 and recovery adds about 0.028564. The query raises the gain-producing quote in an actual fitted model. This witness is not a measurement of human seller behavior. Across 720 primary cells and 720,000 returned-profile rows, the saved $E + R + C = G$ identity is also reconstructed. These checks connect the recovery argument to the empirical model class, but do not identify why feedback training is harmful or estimate the restricted frontier's real-data optimum.

7.2 Recovery-path regularization: improvement and costs

Takeaway. RPL lowers primary mean inflation across all thirty source–seed comparisons at the declared unit-weight endpoint. Predictive, query-failure and convergence costs keep this from being a dominance claim.

The ten-seed extension compared path weights $0, 0.1, 0.5, 1$ at gap weight $\beta = 0.5$, plus path weights $0, 0.5$ at $\beta = 0$, over query capacities $0, 0.1, 0.3$. All branches used no policy feedback and the same two-round reset schedule. The original 162 cells were reused and 378 added, for 540 cells. Each positive path-weight branch is compared with its own same-gap zero-path control. RPL changes the predictor; it is not a fixed-predictor acquisition comparison.

The original grid, including $\lambda = 1$, was committed before formal evaluation in `6643cbe`. This is a timestamped development grid, not independent preregistration or proof that the endpoint was prespecified as the headline. Weight 0.5 retained the existing gap term's numerical scale; neither 0.5 nor 1 is claimed optimal.

Inflation improvement and uncertainty. At 10% capacity and $\lambda = 1, \beta = 0.5$, ratios of mean inflation to the matched $\lambda = 0$ control fell by 50.41%, 26.58% and 23.81% on synthetic, JUCars and MUCars. All ten fitting seeds in each source had lower inflation. The absolute paired reductions were 0.006268, 0.003306 and 0.001573 log units. The exploratory simultaneous bands over 216 metric contrasts were $[0.004644, 0.007891]$, $[0.001209, 0.005403]$ and $[0.000624, 0.002522]$, respectively (Figure 1). These conditional bands support the within-design direction without certifying finite-sample coverage or new-population generalization. Smaller weights were less uniform: at $\lambda = 0.1$, only six of ten seeds improved on each real source; at $\lambda = 0.5$, nine of ten did so. Appendix I retains every seed.

Predictive and verification costs. Table 4 reports absolute outcomes. At the unit endpoint, honest MAE changed by $+0.82\%$, -0.10% and -0.80%, while attacked MAE changed by -0.91%, -0.02% and -0.87%. Failed attempts as a fraction of all listings changed by $+1.10$, $+0.44$ and -0.12 percentage points. The synthetic honest error and failed-query increases also excluded zero in the exploratory simultaneous bands; real-source MAE changes did not. Figure 2 shows all paired seed changes, including unfavorable cases. Its means of seed ratios are distinct from the ratios of means reported above. Inflation improvement alone therefore does not determine a deployment choice.

Budget and loss-component boundaries. Figure 3 includes zero queries, so estimator changes remain separate from gains due to verification. Greater capacity can lower inflation while increasing charged failed queries. The matched path/gap comparison (Figure 4) shows that the path intervention is not an alias for the existing gap term at the tested weights. It complements the original eight-configuration, three-switch component ablation rather than turning its switches into three independent innovations. The original three-seed recovery-edge diagnostic also showed lower positive gain increments, which remained strictly positive; those measurements are retained as a pilot in Appendix I and are not relabeled as ten-seed evidence.

Finite-response boundary. All 540 cells remain in Figure 5. Fifty-one cells contained nonconverged batches, covering 89 of 5,400 batch replays, with a maximum remaining unilateral regret of 0.143721 log units. At the primary unit-weight endpoint, two synthetic seeds each retained one nonconverged batch; the maximum regret there was 0.011042. Both real-source endpoint responses

Table 4: Ten-seed RPL means at 10% capacity, $\beta = 0.5$. MAE is in source currency; failed queries are a percentage of all listings. All seeds and nonconverged responses are retained.

Source / λ	Inflation	Honest MAE	Attacked MAE	Failed (%)
synthetic / 0	0.01243	3576.06	3635.45	2.51
synthetic / 0.5	0.00707	3595.31	3600.02	3.49
synthetic / 1	0.00616	3605.35	3602.26	3.61
jucars / 0	0.01244	6966.89	6957.18	5.73
jucars / 0.5	0.01077	6948.32	6941.48	5.35
jucars / 1	0.00913	6959.75	6956.01	6.17
mucars / 0	0.00661	27155.74	27306.07	0.43
mucars / 0.5	0.00566	26983.89	27114.40	0.39
mucars / 1	0.00503	26938.93	27069.81	0.31

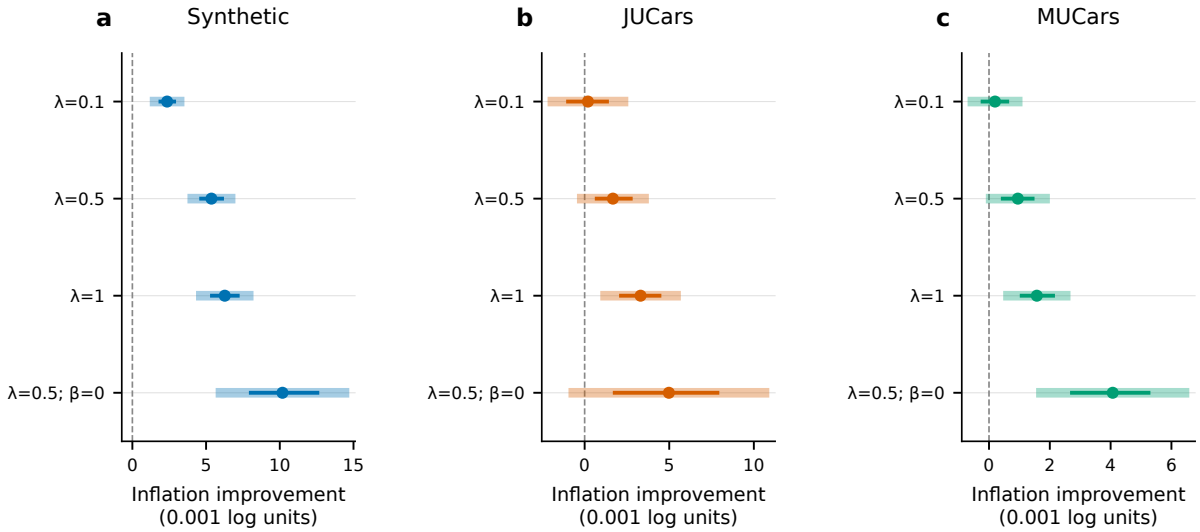


Figure 1: Ten-seed paired inflation improvement at 10% capacity. Positive values favor RPL. Each branch uses its same-gap zero-path control. Thin segments are marginal 95% intervals; pale segments are exploratory simultaneous bands over 216 metric contrasts. Ten fitting seeds share fixed training/evaluation populations and ten batches. Bands do not establish finite-sample family-wise coverage.

met the stated finite unilateral tolerance. These are outcomes of the declared finite response search, not evidence of improvement at an exact market equilibrium.

7.3 Five task-adapted recent acquisition families on fixed price models

Takeaway. These are five task-adapted comparisons, not unchanged reproductions of five original methods. Their four-stage computational checks are complete; estimator choice and simple strong controls remain material.

ACO is a clean-room one-query adaptation. SEFA uses official model code inside a changed task interface; GDFS and DIME use the SEFA repository's comparison implementations; explanation-ranked AFA uses official network classes with adapted ranking generation and training. Origi-

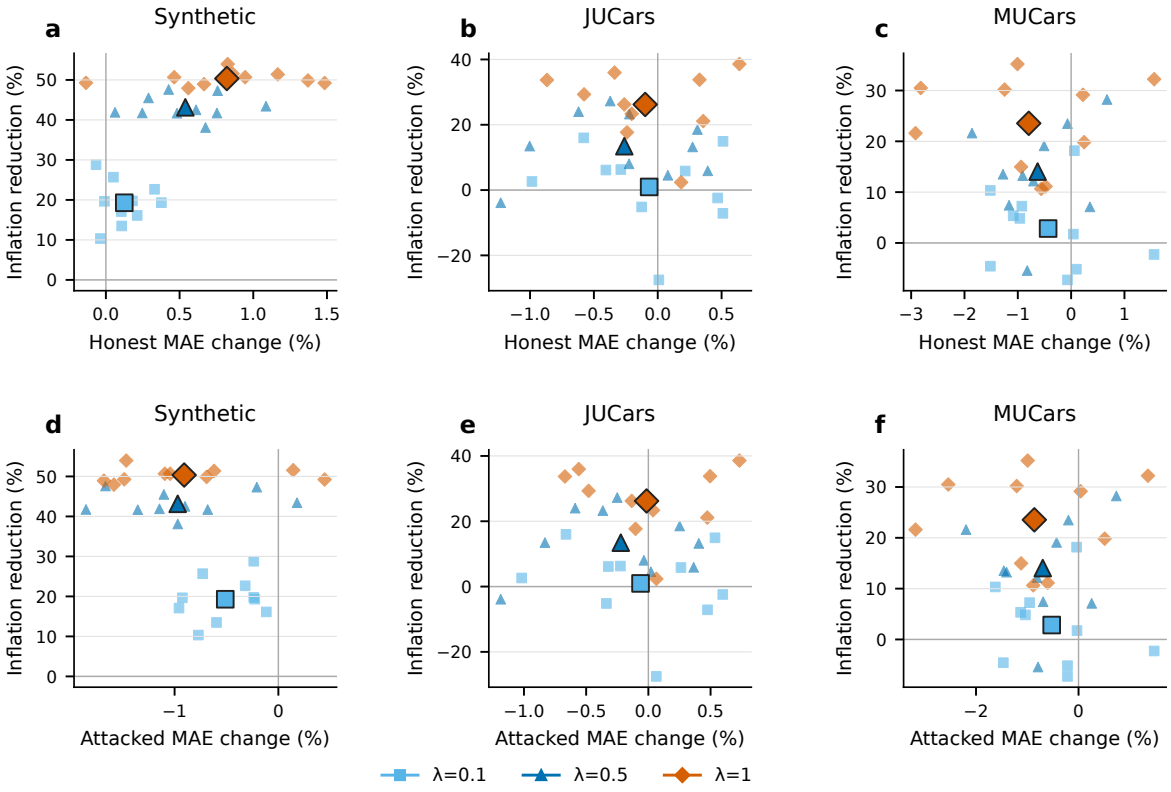


Figure 2: Paired seed trade-offs at 10% capacity and $\beta = 0.5$. Small markers retain all ten fitting seeds per source and positive path weight; outlined markers are their means. Horizontal axes show honest or attacked MAE changes, while vertical axes show inflation reductions. Upper-left is favorable. Each point uses its same-seed control, so these mean seed ratios differ from the headline ratios of means. No confidence region is claimed.

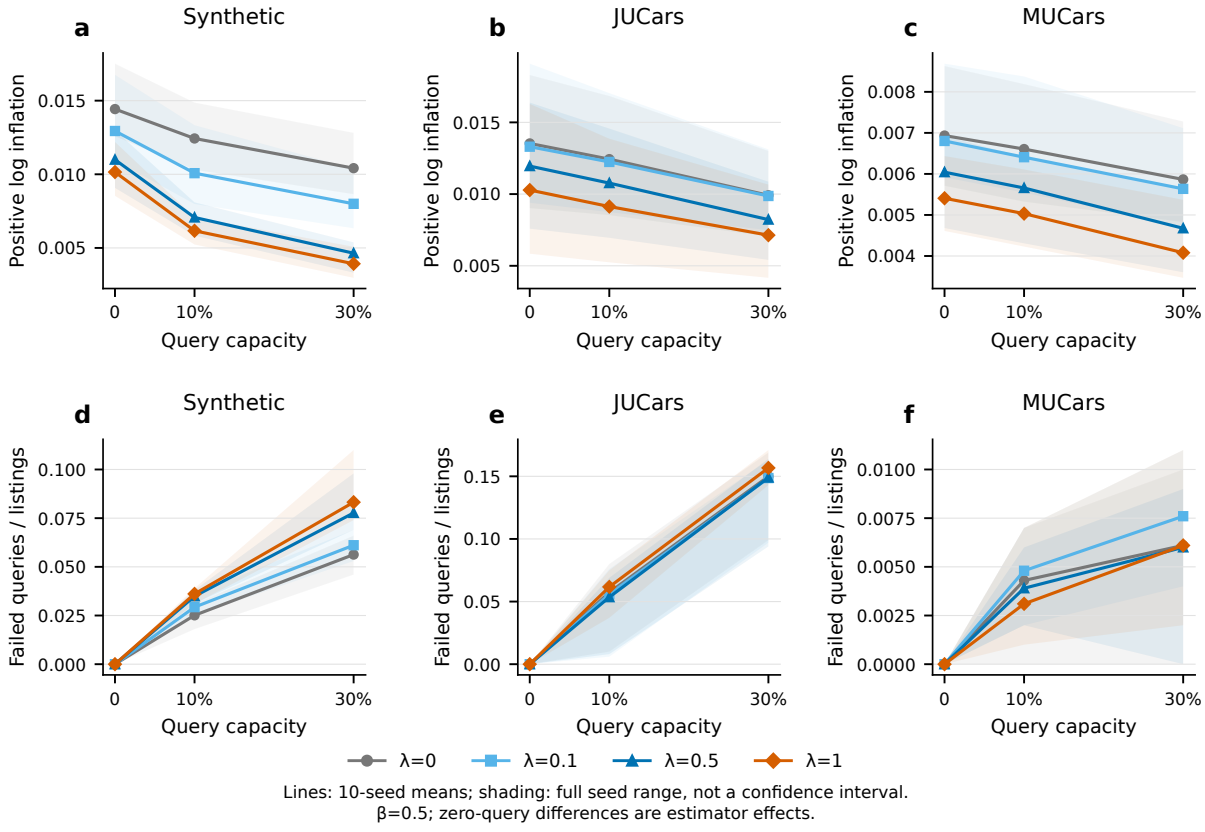


Figure 3: Budget transfer and verification costs at $\beta = 0.5$. The measured capacities are 0, 10 and 30%; straight segments connect evaluated settings only. Lines show ten-seed means and shading the full seed range, not a confidence interval. Failed queries use all listings as the denominator. Zero-query differences are estimator effects.

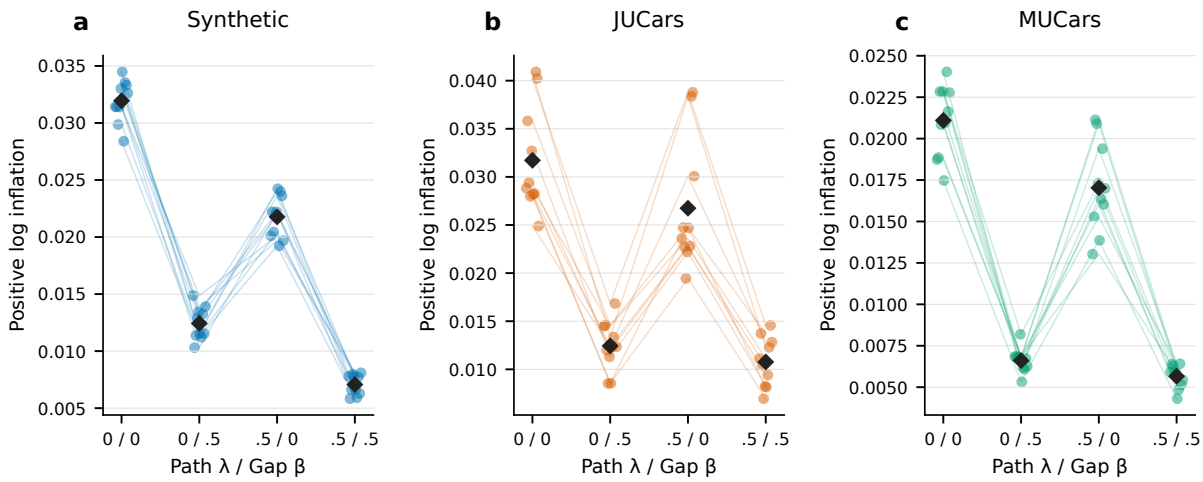


Figure 4: Matched path/gap loss-component comparison at 10% capacity. Each line connects one fitting seed across four configurations; dots retain all observations and black diamonds show means. The axis lists path weight/gap weight. This tests the path term against matched zero-path controls with and without the gap term.

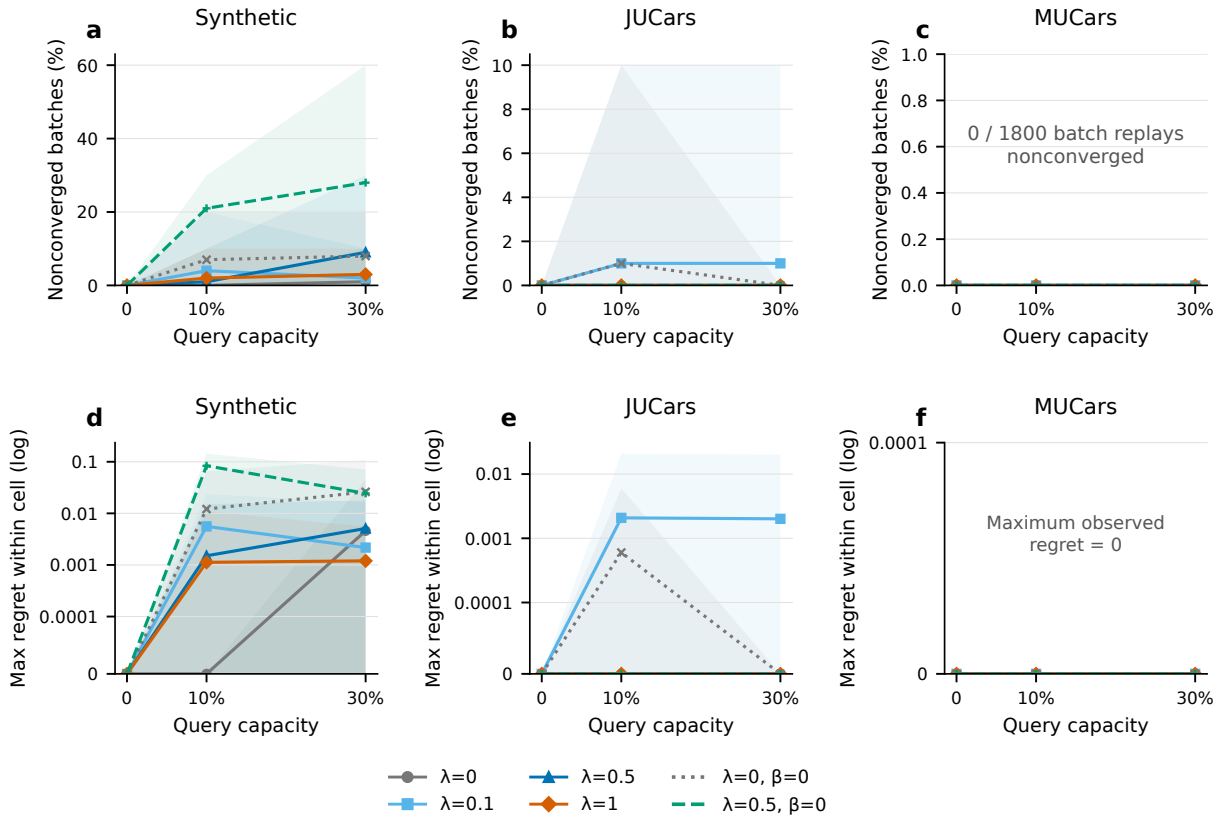


Figure 5: Convergence boundary across all 540 RPL cells and six configurations. Top panels show batch nonconvergence rates; bottom panels show the maximum unilateral regret within each cell on a symmetric-log axis. Curves are ten-seed means, bands the full seed range. The mean of cell maxima is not the worst observed regret. All 51 cells containing nonconverged batches are retained.

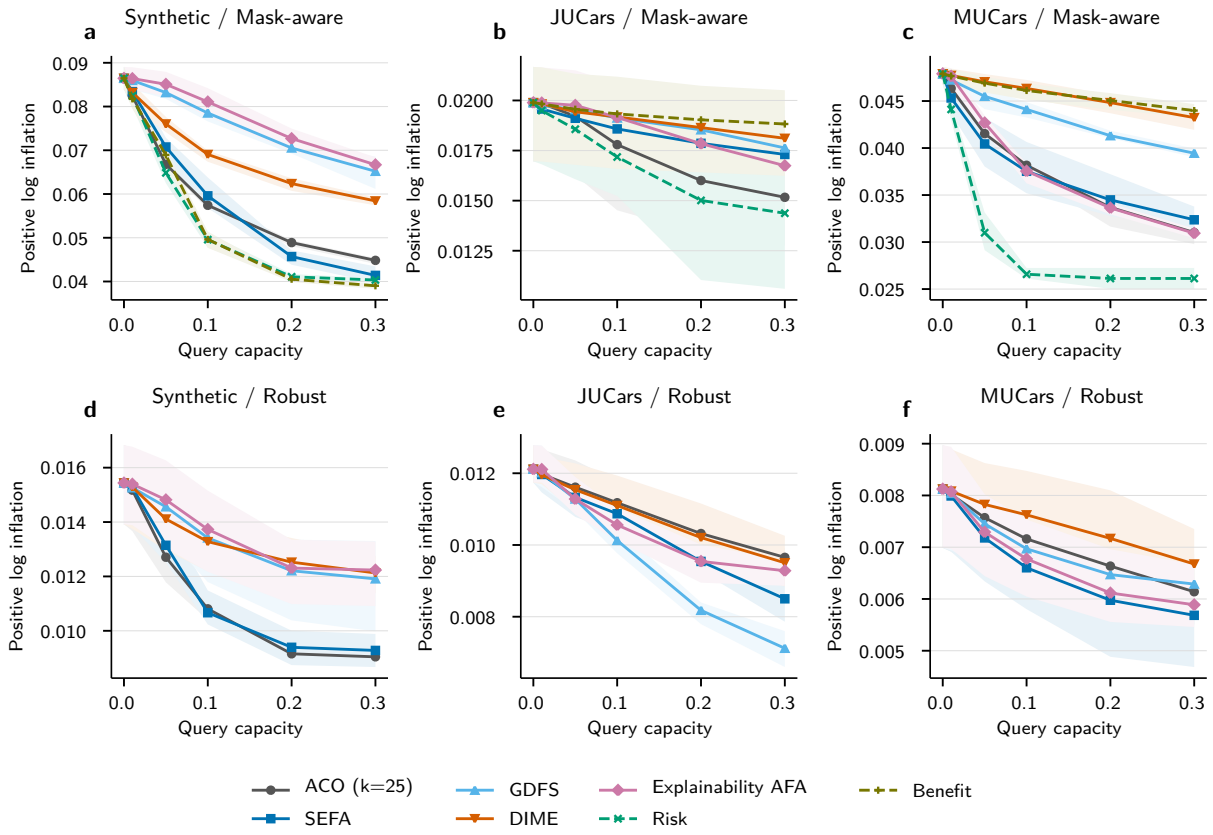


Figure 6: Five recent acquisition families under shared one-query batch capacity. Lines are means over three fitting seeds; shading is the full seed range, not a confidence interval. Rows fix the mask-aware or robust price predictor and columns denote sources. ACO uses fixed $k = 25$. Risk and benefit use their available matching mask-aware backend; no matching robust-backend version is invented. All nonconverged outcomes remain included. The four classification-based adapters use training-only price bins, and every method has explicit task-specific adaptation boundaries.

nal code provenance therefore does not make the complete experiment an official reproduction. The classification-to-regression, observation, selection and budget differences are specified in Appendix H. No claim of reproducing or defeating the original papers follows from this comparison.

Each of the five adapters covered three sources, three fitting seeds, two frozen price backends and six capacities: 540 common-grid cells. ACO additionally retained 36 fixed-neighbor sensitivities. Holding source, seed, backend and capacity fixed, honest MAE agreed across methods to less than 10^{-6}, as required when only acquisition changes. ACO's declared $k = 25$ was used throughout; no evaluation-best neighbor count replaced it. Figure 6 shows every family's budget curve alongside available matched risk/benefit controls. At 10% capacity with the robust backend, GDFS had the lowest family-mean inflation on JUCars (0.01013), whereas SEFA was lowest on MUCars (0.00660) and synthetic (0.01067). No recent family minimized inflation on all three sources. Simple risk or benefit remained competitive or better in several mask-aware comparisons. Full four-metric primary means are in Appendix H; an inflation ranking is not a multiobjective dominance result.

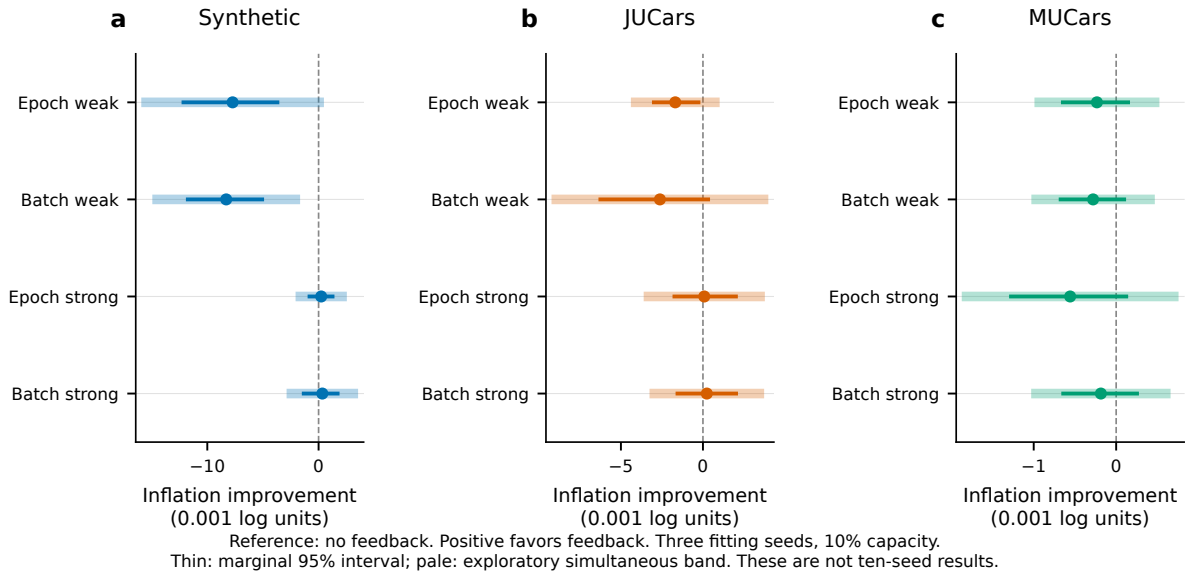


Figure 7: Search-strength and score-refresh interventions relative to matched no-feedback controls, at 10% capacity. Points are mean reference-minus-candidate inflation differences; positive values favor feedback. Thin segments are marginal 95% crossed-bootstrap intervals and pale segments are exploratory simultaneous bands. This intervention retains three fitting seeds and ten shared batches; it is separate from the ten-seed RPL extension.

7.4 Controlled components and a search-strength intervention

Takeaway. Stronger training-response search removed some feedback damage; the original negative result does not describe every feedback implementation.

The complete suppression-score/feedback/gap factorial retained all eight combinations, including removal of two or all three switched components (Appendix H). These switches are experimental components, not three independently established innovations. Gap removal substantially increased mean primary inflation on the real sources, whereas the full three-component configuration was not uniformly best.

The original scoring network was refreshed each epoch, rather than frozen throughout training. We crossed epoch versus minibatch score refresh with weak (two sweeps, one start) versus strong (20 sweeps, four starts) training-response search. Both retained two six-epoch rounds and matched no-feedback references. Figure 7 shows the audited paired improvements and conditional intervals at 10% capacity, retaining three fitting seeds. On synthetic, epoch-weak inflation was 0.02075, epoch-strong 0.01279, batch-strong 0.01269, and no-feedback 0.01303. Thus strengthening search substantially reduced the weak implementation's damage, and the strong means were slightly below the no-feedback reference. Strong JUCars means were also slightly lower than no-feedback, whereas all four MUCars feedback means remained higher. Changing score-refresh frequency alone was not a consistent remedy. These are conditional mean comparisons, not proof of a uniquely identified cause or population-wide feedback benefit.

7.5 Historical feedback and allocation controls

The original factorial had adverse mean inflation directions in all 36 primary feedback contrasts. Under post-hoc exploratory simultaneous families, 31 adverse intervals remained for the inflation-only family, 27 for three co-primary outcomes, and 20 for all 1,536 primary metric contrasts. No family established that every feedback contrast was harmful. These results describe the original weak-search implementation; the stronger-search intervention above changes that interpretation. Appendix K retains all original intervals, interactions and counts.

The strong-control comparisons also restrict claims for the original SAV candidate. Ordinary-benefit and acquisition-conditioned controls had lower inflation in synthetic and JUCars comparisons, and the original ten-seed SAV comparison did not show a universal advantage over static robust valuation. These original-protocol comparisons are distinct from the new same-gap RPL contrasts. Likewise, score-minimax allocation passed the descriptive joint gate in only two of eighteen primary contrasts, while ten worsened inflation. An advantageous interaction or a more complex allocator therefore does not replace a matched no-feedback or simple-acquisition control. Complete three-source evidence is retained in Appendices C and J.

7.6 History-cap outcomes and deployment costs

The history cap's fixed-peer theorem does not imply a better joint outcome after sellers recompute their reports. The 432-cell recomputation kept the original response-search budget, public scores and rule weights. Its multiobjective result is consequential: attacked MAE increased in 94 cells and failed-query rates in 145, despite unchanged honest MAE. Figure 8 shows those costs alongside inflation reductions. Complete per-rule counts and conditional intervals are in Appendix J.

Thirty-three capped cells retained at least one nonconverged batch and maximum remaining regret was 0.204423 log units. No capped result is therefore described as a certified market equilibrium. The descriptive 10% inflation / 5% honest-error gate does not constrain attacked error or failed queries; threshold sensitivity cannot turn a single-metric improvement into Pareto dominance. These costs also motivate the final-record alternative and its natural-missingness boundary in Section 5.

7.7 Applicability beyond the zero-cost capped design

Positive concealment costs changed some rankings through deterrence. In the synthetic top-B subset, stipulated costs near 0.02 sharply reduced concealment and moved outcomes toward ordinary valuation; feedback no longer worsened the mean inflation in the high-cost specifications. Yet over 94% of attempted queries failed in the uniform-cost subset, residual concealment remained positive, and the real-source directions stayed adverse. This supports a conditional regime change rather than a general feedback remedy. The full 108-cell cost design and net-deviation bounds remain in Appendices K and G.

A separate frozen-original-model check removed the real-source evaluation caps, covering all 1,374 JUCars and 10,672 MUCars validation records. All 24 feedback point directions remained adverse. The two MUCars masking/suppression intervals still crossed zero, with fitting-seed variability dominating the uncapped conditional variance. The cohort and record-count weighting both changed, so this is not a pure sample-size experiment or evidence of zero effect. It does not extend the RPL evaluation population: the RPL ten-seed study retains its fixed 1,000 listings per source. Complete uncapped counts, weighting and variance decompositions are in Appendices K and G.

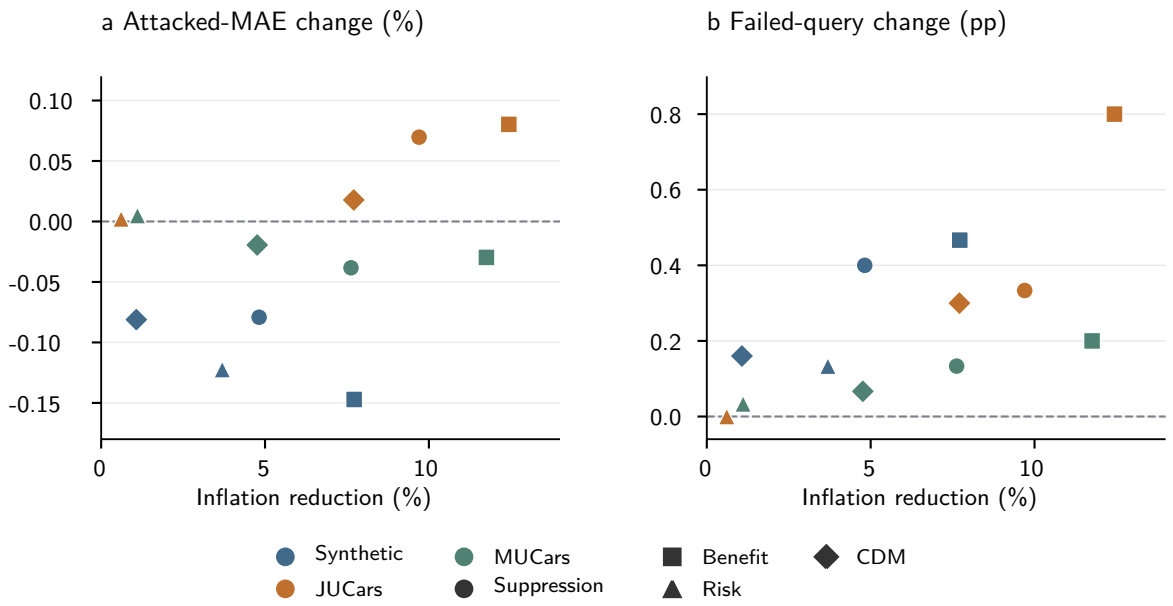


Figure 8: Inflation reductions under the history cap coexist with predictive-error and query-failure costs. Both panels show the same twelve primary robust-estimator source/rule means at 10% capacity, averaging three fixed fitting seeds on 1,000 shared listings per source. Colors denote sources and shapes denote query rules. Positive inflation reduction is favorable; positive attacked-MAE change (a) and failed-query change (b) are unfavorable. Error changes are within-source percentages; failure changes are percentage points per listing. These are descriptive means, without confidence intervals; nonconverged responses remain included. The full numerical comparison is in Appendix J.

8 Discussion

Takeaway. The contribution is a recovery-aware intervention supported by matched ten-seed effects and explicit deployment costs. Verification quality depends on how recovered records enter the predictor and response procedure, as well as which information is acquired.

The information frontier and learned recovery graph address complementary questions. Observationally identical missing reports restrict what a budgeted verifier can guarantee, while recovery paths reveal how a valid query can increase a learned quotation. RPL translates the latter diagnosis into a loss term and reduces inflation in the tested finite-response design. The effect does not require new private deployment inputs, but its zero-query component shows that part of the benefit comes from changing the estimator. Existing predictive acquisition objectives [12] and strategic withholding models [9, 5] therefore remain relevant comparators, rather than being displaced by the recovery-path objective.

Search-strength interventions qualify the earlier negative feedback finding. They show that an adverse weak-search implementation need not characterize all feedback procedures; they do not isolate one unique causal defect or establish feedback benefit across sources. RPL likewise does not dominate all objectives: the synthetic predictor pays higher honest error and failed-query costs. The history cap supplies a restricted fixed-peer guarantee, but its history dependence raises consistency and appeal issues for equal final records. Deployments should choose among these interventions using the joint error, failure and incentive outcomes, not inflation alone.

The scientific scope remains finite and retrospective. Seller concealment, costs, query success and response search are modeled, not observed behavior. The real data are advertised prices from two cohorts, not verified transactions. Fitting seeds quantify algorithmic variability on fixed populations and cannot establish uncertainty over new training populations. Nonconverged response batches bound equilibrium interpretations. Four-stage computational checks improve traceability but share loading and prediction primitives. A separate external run completed, yet failed the original numerical tolerances; this limits cross-machine reproducibility rather than being silently absorbed into the uncertainty bands. Testing learned recovery effects with actual seller decisions and independently controlled numerical environments would address these two remaining transportability questions.

9 Conclusion

Information recovery does not by itself ensure better valuation incentives. A bounded information frontier and observed gain-increasing recovery paths motivate RPL, which lowers primary mean inflation across all thirty tested source–seed comparisons at the declared unit-weight endpoint. Honest-error, query-failure and response-convergence costs prevent a universal dominance claim. Stronger response search also narrows the original negative feedback finding to its implementation. Together, these results support diagnosing recovery paths before choosing feedback or regularization and retaining prediction and verification costs in the deployment decision. The findings remain conditional on the fixed cohorts and modeled finite unilateral responses.

Statements and declarations

Data and code availability. The source datasets are available under CC BY 4.0 through the versioned Mendeley Data records [1, 2]. Prices are advertisements. The public repository, `github.com/T0Q0R/secondhand-valuation-research`, contains project-authored code under MIT, source

reconstruction instructions, aggregate results, figure source data, resampling tensors and audit receipts. Third-party software and data retain their own licenses. Large model weights and per-listing artifacts remain on the research machine; a DOI-backed archival release has not yet been issued.

External execution and numerical reproducibility. An independently operated Mac completed the six-cell synthetic/seed-13 training-and-evaluation package. Training/evaluation populations and tree backbones matched, but residual-network parameters and aggregate outcomes did not meet the original numerical tolerances. Those failures are retained. Subsequent saved-model diagnostics reproduced each machine's own parameters and found the first recorded cross-machine difference in first-batch output weight gradients. Hidden intermediate activations were not captured, so a specific arithmetic primitive or environment was not causally identified. Effect directions at the three tested budgets agreed in this one seed, but that is not numerical equivalence or a multi-seed external confirmation. An isolated float64 engineering candidate has not replaced the reported float32 experiments and has not established cross-machine performance.

Use of AI tools. OpenAI Codex assisted implementation, experiment orchestration, analysis code, literature checks, data-derived figure preparation and manuscript drafting. Figures were plotted from inspectable saved results, not generated as invented empirical images. Quantitative statements are bound to experiment records and the computational checks above. AI tools are not authors; the named authors remain responsible for scientific interpretation and final approval. Author contributions, funding and competing-interest declarations require author confirmation before submission and are not inferred here.

A Contextual policy-library witnesses

A capacity-related global-mixture gap. Consider $n > B \geq 1$, a fixed core C of size B, and public contexts $j \in \{B+1, \ldots, n\}$. In context j, potential quote gaps are 2 for core items, 1 for item j, and $\epsilon = 1/(B+2)$ for every other item. These are linear toy utility units. Use the honest-exact missingness-alias worlds from the preceding construction. All items are query-eligible in the all-missing report.

The reference library contains *all* fixed item-priority orders and a risk rule using these exact gaps, but employs one context-independent mixing vector. Every priority rule selects a fixed B-subset. Risk always selects the core, because $2 > 1 > \epsilon$, without relying on a cutoff tie. Thus the global mixture has the same query marginals in every context, whereas a publicly conditioned mechanism may change them with j.

Proposition 4 (Global-mixture separation)**.** *In this family the optimal worst gains are*

$$T_{\text{global}} = \frac{2(n-B)}{2n-B}, \qquad T_{\text{context}} = \frac{2}{B+2}.$$

Their ratio is $(B+2)(n-B)/(2n-B)$*. Taking* $n = B^2 + B$ *makes the ratio* $B(B+2)/(2B+1)$*, which grows without bound with* B*.*

Proof. A global mixture with value t must satisfy $2(1-p_i) \leq t$ for each core item and $1 - p_i \leq t$ for each other item, using the context in which that item is active. Summing gives $B \geq n - t(B/2 + n - B)$ and the stated lower bound. It is attained with core marginal $1 - T_{\text{global}}/2$ and other marginal $1 - T_{\text{global}}$. These sum to B and are implementable as a lottery over B-subsets, all represented by fixed priorities. In one public context, the B core items and active item imply $B \geq B + 1 - t(B/2 + 1)$. Set core marginal $(B+1)/(B+2)$, active-item marginal $B/(B+2)$ and all others zero to attain the bound. The low-gap unqueried items have gain $\epsilon < T_{\text{context}}$. Both constructions use the preceding hard-budget sampling argument. □

We checked rational primal/lower-bound witnesses on 26 predetermined (n, B) cases and 257,274 context–item rows. Sixteen small cases also match independently constructed complete-subset LPs for both policy classes. At $B = 20, n = 420$, the ratio is $440/41 \approx 10.73$. This is a constructed limitation of global mixing over the specified context-insensitive rule library; it does not establish that real-source CDM losses arise from this mechanism or that these bounds are first in the literature.

A stronger small-library comparator. The separation concerns the information used by the rules, not fixed mixture weights alone. Add $B + 1$ deterministic context-aware rules: one queries every core item; rule i replaces core item i with the publicly active item. Give the first rule weight $2/(B+2)$ and each replacement rule weight $1/(B+2)$. The weights are fixed across contexts. Every draw uses exactly B queries, while each core item's marginal is $(B+1)/(B+2)$ and the active item's marginal is $B/(B+2)$. Thus this small library attains T_{context} without solving a new LP at deployment. The previous lower bound proves its optimality within the constructed family. It does not imply an unseen-context guarantee for learned rules on vehicle records.

An independent exact-arithmetic extension checks all 26 cases, 15,922 context–rule allocations and 257,274 context–item losses. Freezing the active item or changing the weights to uniform fails the optimum check. Consequently, this family supports a representation limitation of the original library, not an intrinsic advantage of dynamically chosen weights or an LP solver. This stronger comparator must be retained in subsequent algorithm evaluations.

B Fixed-profile cap comparisons

Table 5 retains all four rule comparisons for each source.

Table 5: Primary descriptive comparison: robust frozen estimator, 10% capacity. Mean worst unilateral log-quote gain, equally averaging five fixed peer profiles and three seeds. No confidence interval or practical-threshold pass is asserted.

Source	Rule	Original	Capped	Reduction (%)
jucars	benefit	0.010473	0.009736	7.04
jucars	cdm	0.010389	0.010041	3.36
jucars	risk	0.010748	0.010748	0.00
jucars	suppression	0.010546	0.010007	5.11
mucars	benefit	0.007412	0.006486	12.49
mucars	cdm	0.006780	0.006399	5.62
mucars	risk	0.006540	0.006540	0.00
mucars	suppression	0.007607	0.006984	8.18
synthetic	benefit	0.011395	0.010699	6.11
synthetic	cdm	0.011560	0.011244	2.74
synthetic	risk	0.011423	0.011423	0.00
synthetic	suppression	0.012591	0.012085	4.01

C Complete original-protocol ten-seed comparisons

C.1 synthetic

Takeaway. Increasing the fitting-seed count preserves the negative synthetic result against the original static robust comparator. This section uses ten seeds; preceding cap experiments retain three.

We combine original seeds 13, 42 and 2026 with predetermined seeds 101, 211, 307, 401, 503, 601 and 701 under the same original protocol. All ten synthetic checkpoints pass the source-binding, split, model-receipt and recorded-outcome checks: 620 method/capacity cells and 620,000 listing outcomes. These outcome rows are repeated predictions, not independent evaluation examples. The design contains 1,000 unique evaluation listings in ten shared batches of 100, with ten fitting-seed realizations on fixed training records. JUCars and MUCars are reported separately below; the completed union covers all thirty source/seed pairs.

At 10% capacity, SAV mean positive log-quote inflation is 0.021175 versus 0.012084 for static robust without feedback, a 75.23% increase. Table 6 retains all five original contrasts. The benefit-head variant is the original feedback-trained variant; it is not the separately tuned ordinary-benefit benchmark. This extension does not isolate estimator, acquisition and feedback factors, and it must not be presented as the matched factorial experiment.

For each contrast we construct the ten-seed by ten-batch paired difference matrix. We independently resample seed indices and shared batch indices, retaining their crossed grid and within-cell method pairing. These intervals capture empirical fitting-seed and evaluation-batch variation conditional on the training records; they do not capture training-population uncertainty. Only ten batch clusters are available, so nominal coverage is not established. No multiplicity correction or confirmatory hypothesis-test claim is made. A separate checker reconstructs all five contrasts and bootstrap endpoints from raw outcome records using count-weighted resampling, and rejects

Original comparator	Comparator minus SAV	Nominal 95% interval
Benefit head	−0.004328	[−0.007774, −0.001215]
Static robust	−0.009091	[−0.012355, −0.006323]
Masking benefit	0.029153	[0.024928, 0.033582]
SAV without gap	0.017007	[0.014282, 0.019794]
Pre-audit attacker	−0.007556	[−0.010444, −0.004923]

Table 6: Synthetic, original protocol, ten fitting seeds and 10% capacity. Positive paired inflation differences favor SAV. Intervals are exploratory, unadjusted percentile intervals from 1,000 crossed seed/batch resamples.

altered effect direction, interval endpoints and missing-seed inputs.

C.2 JUCars and sensitivity to seed variability

Takeaway. The original-protocol JUCars result does not support a stable SAV advantage, and its uncertainty depends materially on including fitting-seed variation.

Table 7 retains the five primary comparisons. The same ten-seed source-binding and recorded-outcome audit covers 620 JUCars method/capacity cells and 620,000 outcome rows, again representing 1,000 unique evaluation listings in ten shared batches. All five raw-record contrasts and bootstrap endpoints pass independent reconstruction. At 10% capacity, SAV inflation is 0.014988 versus 0.012082 for static robust, a 24.05% increase. The comparator-minus-SAV difference is −0.002906. Its fixed-seed batch-only interval is [−0.003783, −0.002103], whereas the crossed seed/batch interval is [−0.007025, 0.000777]. The latter includes zero. This is uncertainty about a paired effect, not evidence of equivalence or a demonstrated source interaction.

Original comparator	Comparator minus SAV	Nominal 95% interval
Benefit head	0.001252	[−0.000172, 0.002946]
Static robust	−0.002906	[−0.007025, 0.000777]
Masking benefit	0.005030	[0.001649, 0.007985]
SAV without gap	0.019588	[0.016131, 0.023292]
Pre-audit attacker	−0.001037	[−0.002428, 0.000303]

Table 7: JUCars, original protocol, ten fitting seeds, 10% capacity. Positive differences favor SAV. Crossed seed/batch percentile intervals use 1,000 resamples, are exploratory and unadjusted, and condition on fixed training records. The limitations stated for the synthetic extension apply.

C.3 MUCars completes the source comparison

Takeaway. The original-protocol MUCars extension does not establish a stable advantage over static robust or the original benefit head. This comparison differs from the separately tuned ordinary-benefit benchmark.

Table 8 retains the five primary comparisons. The completed three-source union contains thirty source/seed pairs, 1,860 method/capacity cells and 1,860,000 repeated outcome rows. Each source still contains only 1,000 unique evaluation items. A separate raw-record checker reconstructs all fifteen paired contrasts and both bootstrap definitions. At 10% capacity, MUCars SAV inflation is 0.007340 against 0.006511 for static robust, a 12.73% increase. The crossed interval includes zero; this neither establishes superiority nor demonstrates equivalence. Against the original benefit head,

SAV inflation is 3.53% higher, also with an interval spanning zero. The large gains against masking benefit and SAV without gap remain, illustrating why comparator identity must accompany any statement that MUCars favors SAV.

Original comparator	Comparator minus SAV	Nominal 95% interval
Benefit head	−0.000250	[−0.001092, 0.000488]
Static robust	−0.000829	[−0.001870, 0.000210]
Masking benefit	0.039742	[0.036708, 0.042906]
SAV without gap	0.015624	[0.014036, 0.017120]
Pre-audit attacker	0.000580	[−0.000212, 0.001238]

Table 8: MUCars, original protocol, ten fitting seeds, 10% capacity. Positive differences favor SAV. Crossed seed/batch percentile intervals use 1,000 resamples and retain the fixed-training-record, exploratory and unadjusted interpretation of the other two source extensions.

D Conditional probability-stability diagnostics

Takeaway. Randomized allocation admits a conditional sensitivity bound, but small field margins and scores near zero limit its applicability. Stability alone does not imply lower strategic inflation.

Fix the positive-score support I, the queried field of each item, and budget B. For two score vectors g, g' with all supported scores at least $m > 0$, let $\delta = \|g - g'\|_\infty$. Define the score-minimax value and canonical marginals

$$T(g) = \min_{0 \le p \le 1,\, \sum_i p_i \le B} \max_{i \in I} g_i(1 - p_i), \qquad p_i(g) = [1 - T(g)/g_i]_+.$$

For $B = 0$ use zero marginals; for $B \ge |I|$ use unit marginals on I. Then

$$|T(g) - T(g')| \le \delta, \qquad \|p(g) - p(g')\|_\infty \le \min\{1, 2\delta/m\}.$$

For any fixed feasible allocation, the two objectives differ by at most δ, which proves the value bound after minimization. The function $h(t, x) = [1 - t/x]_+$ is separately $1/m$-Lipschitz in t and x for $t \ge 0, x \ge m$: on its positive branch, $t/x^2 \le 1/x$, and it is continuous at the branch boundary. Applying the value bound proves the marginal bound. The active allocation set may change; positive-score support and queried fields must remain fixed.

If both pre-query and recovered quotes change by at most ϵ, and the primed recovery span is at most R, expected quotes differ by at most $\epsilon + R \min\{1, 2\delta/m\}$. If these conditions hold uniformly over all own reports at fixed peer reports, costs are unchanged, and truthful quotes are verification-neutral, the positive unilateral-gain envelope changes by at most $2\epsilon + R \min\{1, 2\delta/m\}$. This is not a guarantee for adapted peer responses, equilibrium selection, training convergence, or a fixed realization of the sampling randomness.

Three examples delimit the claim. With two items and $B = 1$, swapping scores $(1 + \delta, 1)$ and $(1, 1 + \delta)$ changes the first item's top-B probability by one, but its minimax probability by $\delta/(2 + \delta)$. In contrast, swapping $(2\delta, \delta)$ and $(\delta, 2\delta)$ changes its minimax probability by $1/3$ even as δ tends to zero. Finally, one fully queried item with two fields whose scores swap as in the first example can have a unit quote jump if the two recovery quotes are zero and one. The score lower bound and fixed-field condition therefore cannot simply be removed. An exact-arithmetic checker covers 2,916 score-pair/budget cases, including 510 active-set changes; 108 production/linear-program comparisons agree to floating-point precision. These finite checks supplement the proof.

We additionally audited all 1,080 saved fixed-profile batch executions at hypothetical radii $\delta = \alpha\,\mathrm{median}(g_i : g_i > 0)$. A sufficient certificate requires support preservation and every positive item's top-two field gap to exceed 2δ. Its bound uses $m = \min_{i \in I} g_i - \delta$. Recovery spans are reconstructed for every eligible item, including items not queried by a saved top-B allocation. Table 9 records the sufficient-certificate counts. Each row has 1,080 batch executions, including both

α	Minimax conditions	Probability bound < 1	Top-B constant
0.001	471	468	401
0.01	401	353	212
0.05	267	245	53
0.1	244	242	39

Table 9: Post-hoc stability certificates at four hypothetical relative score radii. Each row covers the same 1,080 dependent fixed-profile batch executions. Counts certify sufficient conditions, rather than observed perturbation errors or a ranking of performance.

generating allocators; these are dependent diagnostic cases, not independent statistical replicates. Top-B is locally constant when the same support/field conditions and its selection-rank margin hold (or all positive items fit). Its certificate is stronger than a nonzero sensitivity bound, so counts do not establish an ordering of stability. Failure to certify is not evidence of instability. A separate checker reconstructs all 4,320 condition rows and 144 grouped summaries using exact fractions of the stored scores and radii; displayed quote bounds remain floating-point approximations. The radii are not calibrated prediction errors. These post-hoc diagnostics neither establish performance improvement nor explain source effects causally. General water-filling sensitivity is established literature; priority for this application-specific bound is not claimed.

E Strong additional comparators and adaptation boundaries

The following controls belong to earlier separately corroborated experiments, not the newly trained matched factorial model family. Keeping these identities separate prevents a favorable acquisition or tuning result from being attributed to the wrong training intervention.

E.1 Ordinary-benefit and equal-allowance controls

A robust/no-feedback price model with ordinary error-reduction acquisition was evaluated on all six budgets and the original three seeds. The independently checked 54-cell extension retained identical honest quotes, public scoring, allocations and first-batch exact deviations. At 10% capacity, reference inflation was 0.01077, 0.01117 and 0.00778 on synthetic, JUCars and MUCars, respectively. Reference-minus-SAV differences were -0.00998, -0.00432 and 0.00089, with conditional shared-batch intervals $[-0.01260, -0.00791]$, $[-0.00598, -0.00285]$ and $[0.00045, 0.00130]$. These intervals averaged the three fitted seeds before resampling batches; they are not the ten-seed crossed intervals in the main factorial analysis.

An additional size/tuning study retained training caps 1,000, 2,500 and 5,000, three fitting seeds, three training-internal gap-weight trials and twelve residual epochs for both candidates. Its 972 parent tuning cells and 324 ordinary-benefit reference cells passed full audits. At the largest cap, tuned SAV inflation was 20.41% lower on MUCars but 30.09% higher on synthetic and 27.09% higher on JUCars than the separately tuned ordinary-benefit reference. The associated reference-minus-SAV intervals were $[0.00097, 0.00186]$, $[-0.00682, -0.00204]$ and $[-0.00333, -0.00125]$. All

scheduled caps and fixed/tuned roles remain in the archived evidence; no favorable training size replaces the largest-cap comparison. This further supports conditional gains rather than a consistent candidate advantage.

E.2 Categorical support-margin imputation

A whole-field adaptation of support-margin imputation (SMI) [9] used all allowed covariates and four optional-field groups. One deterministic fitting generation per source trained a sixteen-model linear support-vector regression family on complete-optional-field training rows. Scaling, category vocabularies and target normalization were training-only. Missing categories required a one-hot completion including an unknown token. If the reduced sign-consistent model family or its optimization was unavailable, the method used ordinary median/mode imputation. The original SMI guarantee is not asserted on these fallback rows, and this is not an unchanged official reproduction.

An independent categorical-enumeration and scalar-interval auditor checked 94,527 report/domain cases and every final quotation and legal no-query report. Training cohort sizes were 2,947, 2,674 and 3,368 for synthetic, JUCars and MUCars. Across unrestricted, nonnegative and unit-box completion domains, aggregate outcomes matched ordinary median/mode regression. Attacked fallback fractions were 75.3%, 87.3% and 65.6%, respectively. This diagnoses a limit of the categorical adaptation's retained-family assumptions; it does not refute the original SMI theorem. These no-query fits differ in cohort and model class from the matched factorial predictor, so their higher inflation cannot isolate a benefit of the present acquisition procedure.

E.3 Acquisition-conditioned neighbor control

ACO-1 adapted acquisition-conditioned expected-loss scoring [12] to one-field regression queries and batch capacity. A training-only neighbor bank estimated loss changes for sixteen observation patterns; unavailable recorded bank values gave zero replay benefit. Distances used observed query groups and permitted nonoptional features, with fixed category/numeric weights and stable ties. It was a clean-room task adaptation, not an unchanged execution of the original sequential classification method.

The separately corroborated nine-pair study contained 144 cells. The declared main $k = 25$ used every capacity; fixed $k = 10$ and $k = 50$ sensitivities used 10% capacity only. No winning k was selected. With the frozen robust predictor and $k = 25$, mean inflation was 0.01081 on synthetic, 0.01118 on JUCars and 0.00716 on MUCars. Full-SAV-minus-ACO differences of 0.00995 and 0.00431 on synthetic and JUCars had conditional intervals $[0.00725, 0.01345]$ and $[0.00291, 0.00588]$. On MUCars the point comparison favored SAV by 0.00027, but the interval crossed zero. Thus a strong predictive acquisition control can outperform the original candidate. These results strengthen comparator coverage without implying that one-step ACO-1 dominates the matched factorial policies or reproduces the original method's guarantee.

F Detailed claim–evidence map

Table 10 connects the principal claims to their evidence.

Table 10: Claim–evidence matrix. Positive comparator-minus-SAV differences favor SAV. All listed experiments are complete; bounded claims remain separate from general superiority.

Claim	Decisive evidence	Verdict and boundary
C1. A precision–incentive frontier is tight.	Restricted missingness-alias proof and 32-case finite oracle.	Supported in the stated family; uniform honest error and verification-neutral quotes required.
C2. Marginals replace complete allocation lotteries.	57 compact cases; 56 comparisons with complete enumeration.	Supported for item-local utility, unit costs and at most one field per item; marginal optimization is established prior work.
C3. The fixed-priority library can have an unbounded gap.	26 exact cases; a fixed mixture of $B+1$ context-aware rules closes the constructed gap.	Supports a restriction on rule information; does not establish an advantage of dynamic weights or the LP solver.
C4. A verification-history cap reduces fixed-profile incentives.	Pointwise replay argument and exhaustive fixed-profile evaluation.	Supported for fixed peer reports and the stated truthful replay; not an equilibrium theorem.
C5. The cap improves every measured objective after recomputed responses.	432 cells: 94 attacked-error increases and 145 failed-query increases.	Not supported. Inflation reductions do not imply Pareto dominance.
C6. SAV consistently beats static robust at 10% capacity.	Ten-seed paired difference: synthetic -0.009091; JUCars -0.002906; MUCars -0.000829.	Not supported. Synthetic is unfavorable with an interval excluding zero; both real-source intervals include zero.
C7. Score-minimax allocation consistently improves outcomes.	108 audited cells; 18 independently checked contrasts; only two pass the point gate, while ten worsen inflation.	Not supported generally. MUCars robust gains coexist with failed-query increases; the original factorial does not show beneficial weak-search feedback; stronger-search means qualify that result.
C8. Matched feedback training reduces inflation.	4,320 audited cells; at 10% capacity, all 36 zero-cost feedback points worsen inflation; 31 adverse bands survive the post-hoc 36-contrast sensitivity.	Not supported for this training procedure. Positive costs reverse one subset ranking alongside strong concealment deterrence. Neither a universal feedback failure nor confirmatory coverage is established.
C9. RPL lowers primary inflation at the unit endpoint.	Ten fitting seeds per source; all 30 matched directions favorable, with positive exploratory simultaneous lower bounds.	Supported within the fixed finite-response design; synthetic honest error and failed queries increase, and nonconverged responses remain.
C10. Core extensions have completed computational checks.	Four-stage chain for 1,206 distinct cells, including all 540 RPL extension cells.	Shared primitives and retrospective cohorts remain; external exact numerical reproduction failed.

G Conditional uncertainty and mechanism diagnostics

G.1 Exact conditional crossed-bootstrap variance

Let Y be a paired-difference matrix with S fitting seeds and K batches. Independent bootstrap seed weights U and record-normalized batch weights V give $U^\top YV$. Write $m = \mathbb{E}[V]$, $C = \mathrm{Cov}(V)$, $r = Ym - \mathbf{1}\,\mathrm{mean}(Ym)$, $z = \mathrm{mean}_s(Y)$ and $R = Y - \mathbf{1}z^\top - r\mathbf{1}^\top$. The conditional variance decomposes as

$$\frac{\|r\|^2}{S^2} + z^\top Cz + \frac{1}{S^2}\sum_s R_s^\top CR_s.$$

These are seed-main, batch-main and interaction terms. Batch-weight moments are calculated by conditioning on the multinomial count of the single partial batch, retaining ratio normalization. Exhaustive balanced and unequally weighted 2×2 resampling independently checks the formula. All 48 capped/full real-source feedback decompositions are retained. Capped components use ten equal 100-record batches; uncapped MUCars components use 107 batches normalized by record counts, including a 72-record tail. Thus the outcome matrix and averaging weights both change. For top-B and minimax, batch variances fall by factors 5.73 and 8.04 (not 10.7), while seed-main variances change by -13.20% and -9.43%. This is not variance reduction from sample size alone. Holding all 107 batches fixed and changing only equal-batch to record-count weights changes the seed term by merely -0.16% and $+0.11\%$, respectively; the larger change cannot be attributed to tail weighting alone.

Thirty-seed projections divide seed and interaction terms by three; the batch-noise-free projection holds the empirical batch distribution fixed. Neither supplies new training runs, equivalence evidence or population coverage.

G.2 Concealment identification from saved utility summaries

For one listing let $C_i = c_i k_i$ be concealment cost, G_i positive expected quote inflation, N_i the positive best-legal-utility gain over truth, and ρ_i remaining unilateral regret. Since truthful utility is its honest quote,

$$C_i - \rho_i \le G_i - N_i \le C_i.$$

Thus $\max(\overline{G} - \overline{N}, 0)$ bounds mean cost below; adding $\max_i \rho_i$ and capping by $2\overline{c}$ bounds it above. Uniform-cost hidden-field bounds divide by c; heterogeneous costs in $[.01, .03)$ give bounds using .03 and .01, respectively. Dividing by mean heterogeneous cost would not identify mean hidden fields. Positive-cost states were not saved; all 20 nonconverged batches remain in the bounds. The inequality passes 768 exhaustive toy utility/returned-state checks. The paired zero-cost concealment counts use saved states directly. Neither co-occurrence nor this accounting identity identifies a causal mediation effect.

G.3 Restricted final-record honest-error pilot

Three frozen robust/no-feedback models per real source (seeds 13, 42 and 2026) enumerate completions of transmission, fuel and body condition from their training-encoder vocabularies, including missing tokens. Mileage must be observed and fixed; other covariates stay fixed. Unsupported observed categories and missing mileage abstain. The same original 1,000 records per source are used without tuning or retraining. Current records are included in the minima; singleton sets preserve their original quotes exactly. Shared-support errors and abstentions are reported separately.

Figure 9 links the coverage boundary to the concentration of error costs.

Source	Support	Records	Honest-MAE change	Mean quote reduction
JUCars	Covered	801	0	0
JUCars	Missing category	1	0	0
MUCars	Covered	999	10.49–23.87	200.21–240.51
MUCars	Missing category	28	374.24–851.51	7,143.33–8,581.21

Table 11: Final-record categorical-envelope pilot. Entries are per-source three-seed ranges, not uncertainty intervals; positive MAE changes are worse. Price quantities use source currency units. Missing-category rows are subsets of covered rows, not additional observations. JUCars abstains on 199 records, MUCars on one, all for missing mileage.

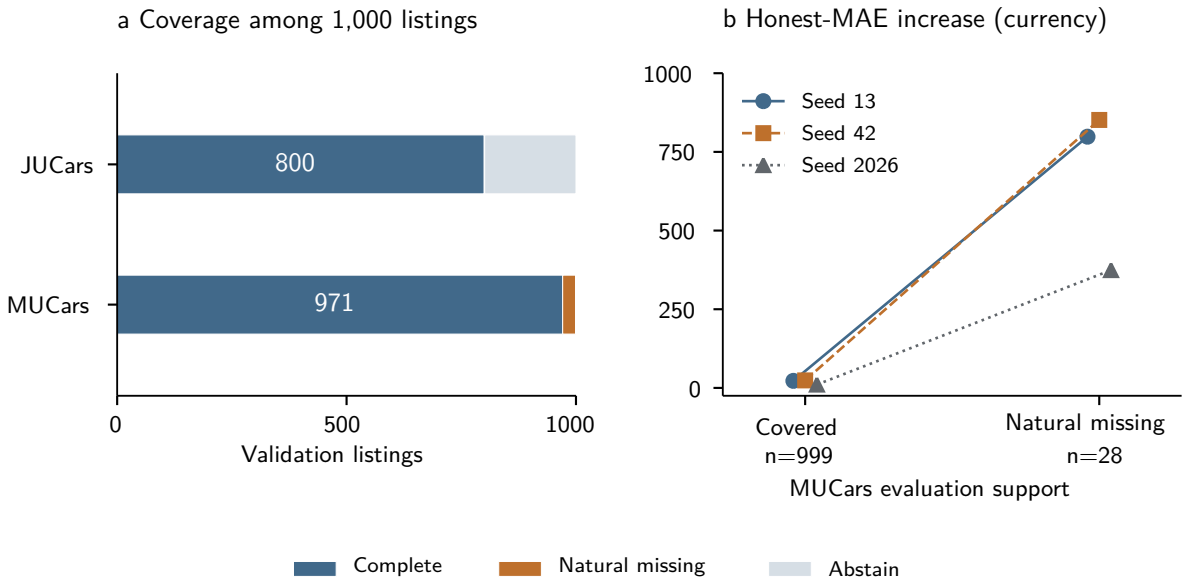


Figure 9: The restricted final-record envelope's honest-error cost is concentrated on naturally incomplete categorical records. (a) Among each source's original 1,000 listings, complete/incomplete/abstained counts are 800/1/199 for JUCars and 971/28/1 for MUCars. Complete means all three declared categorical fields are observed; missing mileage causes abstention. Complete covered records have singleton completion sets and exactly zero change. (b) Every MUCars fitting seed is shown for all 999 covered records and its nested 28-record naturally incomplete subset; lines link the two descriptive means for the same frozen model, without interpolation or confidence intervals. Positive honest-MAE change is worse; units are MUCars currency. Mileage remains observed and fixed. This does not validate the full four-field game or unseen-category coverage.

Each JUCars model evaluates 806 completions; each MUCars model evaluates 1,139. All baseline quotes match saved predictions within 5×10^{-6} in log quote; sampled completions also match the production all-state inference route within that tolerance. Predictor primitives are shared. On the declared covered domain, the envelope is exactly costless for complete categorical records: 800/801 JUCars and 971/999 MUCars records have singleton completion sets. All quotation and error changes are confined to the one and 28 naturally incomplete records, respectively. For the MUCars subset, the largest individual quotation reduction is 25.5% of that record's original predicted quote, while mean honest-MAE increases range from 374.24 to 851.51 currency units (Table 11). Quotation discounts and absolute-error costs have different denominators and must not be conflated. The practical trade-off is zero change on complete covered categorical records but potentially substantial discounts on naturally incomplete ones. The guarantee covers only the declared categorical domain with fixed observed mileage. It does not cover mileage concealment, unseen values or the original full four-field game, and this pilot supplies no confirmatory or universal method advantage.

H Controlled completion protocols and full comparison evidence

H.1 Recent-method identity and training boundaries

The adapters use the same fixed price backends as ACO-1, not each paper's original predictive task. SEFA, GDFS, DIME and explanation-ranked AFA learn acquisition using four training-quantile price classes; their scores feed a shared hard-capacity, one-query allocator. This does not make them native strategic regressors. Nonquery context is available, natural absence is unrecoverable, and deployment scores do not observe a hidden value or its missingness cause. Hidden-mileage perturbation checks left the corresponding public-state scores unchanged. Predictive information scores and policy probabilities are used as cross-listing priority proxies without a theorem of cross-listing cardinal comparability.

SEFA used its official MIT implementation, commit `3af8da2`, configurations 1/5/9, 120 epochs per trial and their original latent-sampling settings. GDFS and DIME used the comparison implementations in that repository, retaining their relaxed-one-hot/temperature and entropy-scaled value/epsilon-greedy cores. Their configurations 1/5/9 use 80 predictor-pretraining epochs and 15 epochs per annealing stage. The task adapter starts from legal public reports and trains at most two recoveries, uses Adam and stagewise best validation states, and does not reproduce the original learning-rate scheduler. This is a sourced task adaptation, not an unchanged original-author reproduction.

Explanation-ranked AFA used the official repository commit `d08f39a`, its tabular MLP and three-layer, four-head GPT, hidden width 128 and embedding width 64. A fitting-partition 250-tree CatBoost classifier supplies TreeSHAP rankings; field-value and missing-indicator absolute contributions for the training label are summed. Natural missingness and numeric category coding are task adaptations. The stages comprise 250 masked-predictor, 200 ranking-supervised and 16 policy-trajectory epochs, with at most two recoveries per training trajectory and a fixed initial action marker at history-free deployment. Learning rates $0.001, 0.0005, 0.0003$ form a fixed local grid; the last is an additional sensitivity value rather than an original default.

All four classification adapters select among three configurations using the same training-only 80:20 split and then refit on all training records. SEFA uses its masked validation NLL and selected epoch, while the other three use mean pre/post-query NLL and their full fixed schedule. Thus selection effort is documented but their validation objectives are not identical. No evaluation outcome is used to select a configuration. ACO-1 instead retains the declared training-bank rule and fixed neighbor sensitivities described in Appendix E. These differences, the fixed 1,000-record completion cohort, and three fitting seeds bound cross-family conclusions.

H.2 Full component removals and parameter choice

S switches suppression rather than ordinary-benefit score training, F switches feedback, and G switches the gap penalty. All 2^3 combinations share the role separation, epoch budget and reset schedule. The conventional backbone, mask augmentation, prediction/attack-error and anchor terms remain when all three switches are removed. Figure 10 therefore is a complete component ablation, not a comparison with an untrained or native predictor.

Core sensitivity varied gap weight over $0, 0.1, 0.25, 0.5, 1, 2$, tolerance over $0, 0.01, 0.02, 0.05, 0.1$, and one/two/four reset-feedback rounds at fixed total 12 residual epochs. The design values 0.5 and 0.02 are fixed engineering choices on the log-quotation scale, not optimal values estimated from these curves. Increases in gap weight reduce inflation means in this grid, but inflation alone cannot select a default. Figure 11 retains both evaluated capacities. A further 144 cells vary anchor weight, attack-loss weight, learning rate, epochs, action-head tree count and hidden width one at a time

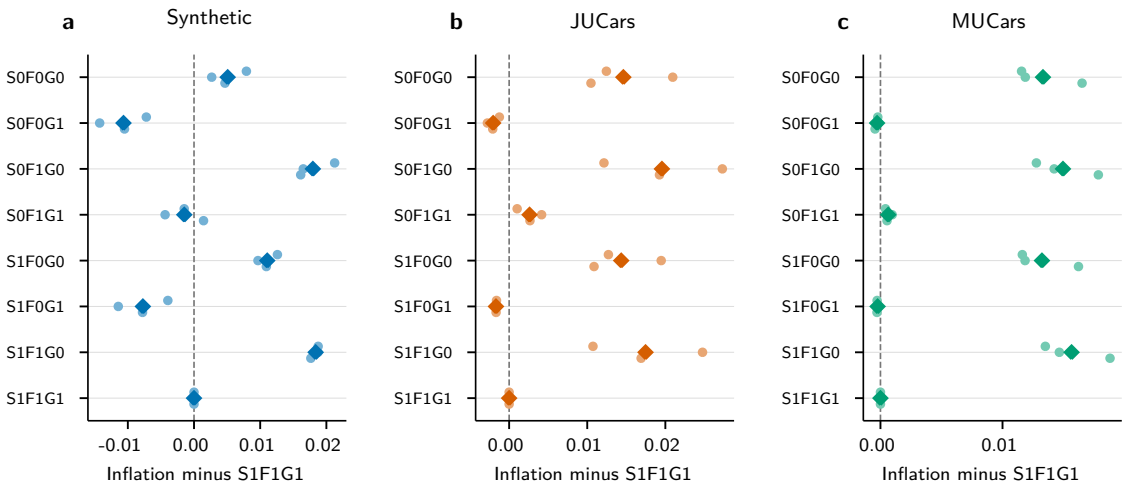


Figure 10: Complete three-component ablation at 10% capacity. Circles retain three paired seed differences from S1F1G1 and diamonds show their means. Negative differences indicate lower inflation than the full switched configuration. S/F/G denote suppression-score, feedback and gap switches. Neither number of cells nor number of components is an independent sample size or an innovation count. All nonconverged responses remain in the plot.

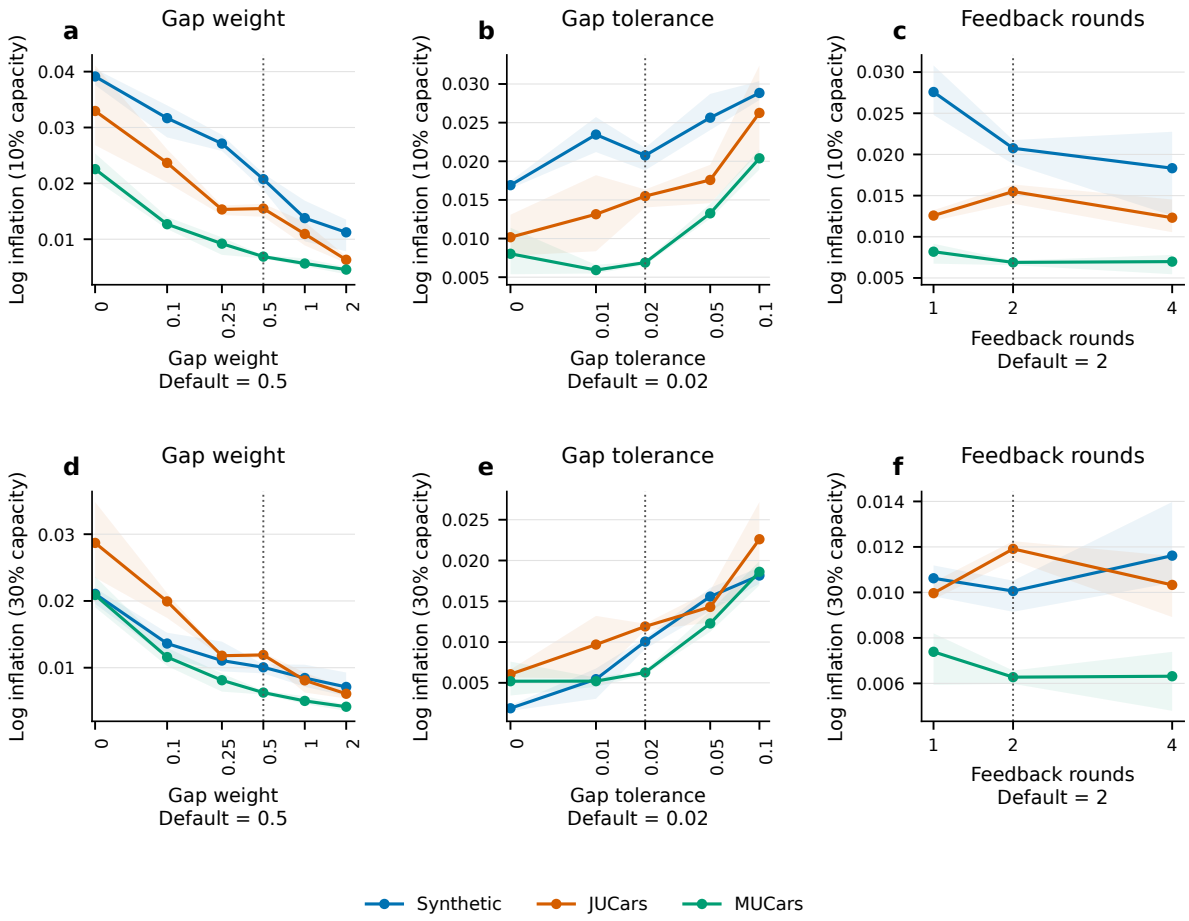


Figure 11: Core-parameter scans in the original feedback implementation, with three fitting seeds. Dotted lines mark design defaults and all evaluated settings are ticked. Gap weight and tolerance use symmetric-log coordinates. Means and full ranges are descriptive, not confidence intervals. The defaults are engineering choices, not evaluation-selected optima.

(Figure 12). The defaults are inserted from the exactly matched S1F1G1 branch, not retrained or interpolated. All error, failure and convergence metrics remain in the source tables.

H.3 Availability-shift stress and source-data coverage

The availability stress study held fitted models fixed and additionally removed 0, 10% or 30% of originally observed query fields using nested stable hashes of listing and field. Targets were unchanged; newly missing values could not be recovered. Seven mechanisms (four search/refresh variants, matched no-feedback, risk and benefit) yielded 189 cells: 63 reused and 126 new evaluations, with no refitting. Reports were reoptimized after the shift. This is a controlled availability perturbation, not a new market, MNAR model or actual-seller validation. Paired metrics, recovery paths and all three seed values remain in the evidence files. RPL additionally retained low/middle/high

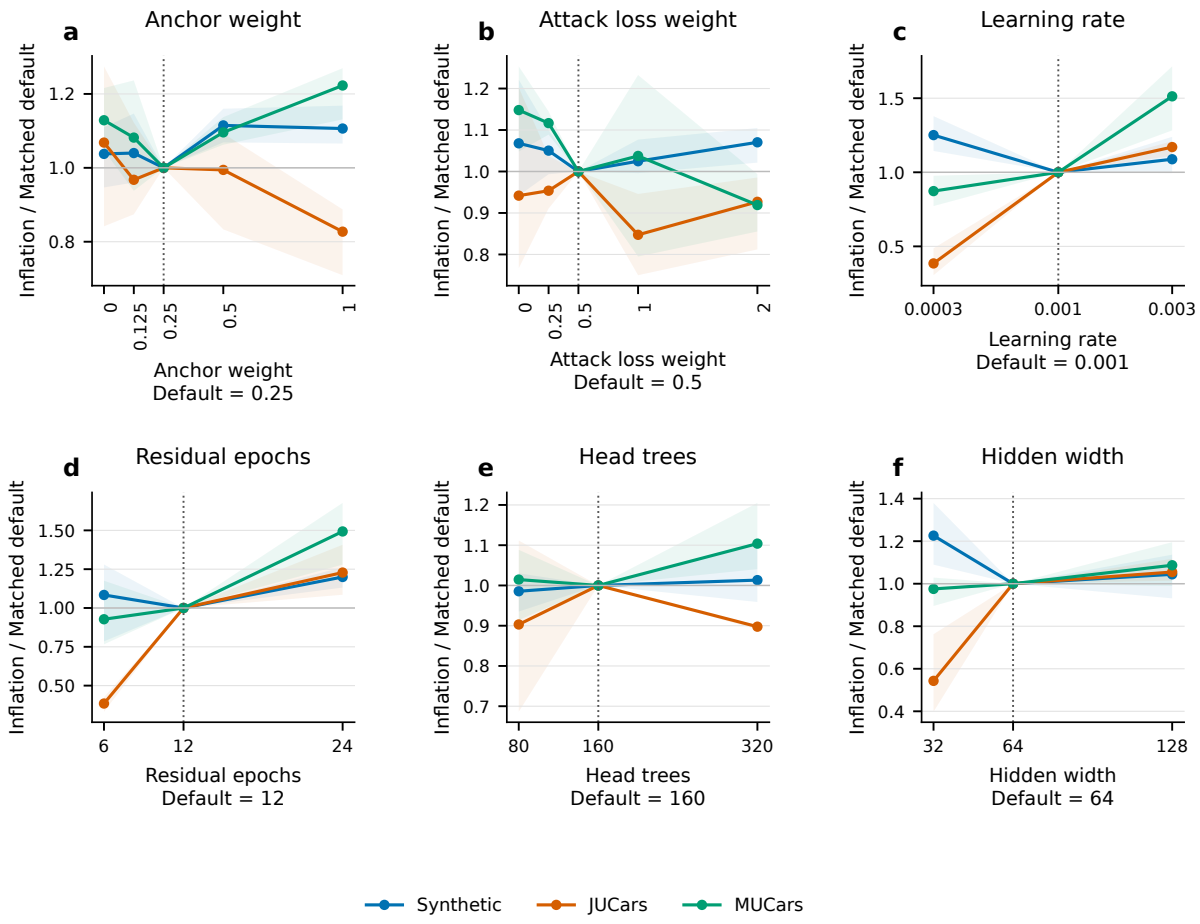


Figure 12: Six implementation-parameter scans at 10% capacity, retaining three fitting seeds. Every ratio uses the same-seed matched default. Dotted lines mark defaults; ticks show all evaluated values, and learning rate uses a logarithmic axis. Curves show means and full seed ranges, not confidence intervals.

price groups using training-only 10th/90th quantiles and naturally complete/missing groups; these shared-population subgroups are descriptive.

H.4 Primary recent-family numerical comparison

Table 12 reports all five family means rather than only winners. Currency MAE is not pooled across sources. Identical honest MAE within a frozen backend is expected and does not demonstrate equal attacked error.

Table 12: Five recent families at 10% capacity. Means of three fitting seeds; M/R denote fixed mask-aware/robust price backends. Failed queries are percent of all listings. ACO uses $k = 25$. Adapter boundaries are given above.

Source	Backend	Family	Inflation	Honest MAE	Attacked MAE	Failed (%)
synthetic	M	ACO	0.05742	3504	4416	3.07
synthetic	M	DIME	0.06905	3504	4734	2.17
synthetic	M	Expl.-AFA	0.08106	3504	5081	2.00
synthetic	M	GDFS	0.07852	3504	5002	1.90
synthetic	M	SEFA	0.05958	3504	4452	2.67
synthetic	R	ACO	0.01081	3568	3612	3.43
synthetic	R	DIME	0.01328	3568	3627	2.37
synthetic	R	Expl.-AFA	0.01373	3568	3644	3.30
synthetic	R	GDFS	0.01341	3568	3640	3.10
synthetic	R	SEFA	0.01067	3568	3602	3.40
jucars	M	ACO	0.01779	6875	6882	0.33
jucars	M	DIME	0.01918	6875	6881	2.47
jucars	M	Expl.-AFA	0.01915	6875	6879	0.40
jucars	M	GDFS	0.01912	6875	6880	3.50
jucars	M	SEFA	0.01858	6875	6878	2.30
jucars	R	ACO	0.01118	7082	7081	3.07
jucars	R	DIME	0.01110	7082	7078	4.17
jucars	R	Expl.-AFA	0.01056	7082	7081	4.43
jucars	R	GDFS	0.01013	7082	7079	4.83
jucars	R	SEFA	0.01087	7082	7081	4.37
mucars	M	ACO	0.03817	27282	28083	0.60
mucars	M	DIME	0.04634	27282	28327	0.20
mucars	M	Expl.-AFA	0.03755	27282	28144	0.00
mucars	M	GDFS	0.04411	27282	28246	0.33
mucars	M	SEFA	0.03753	27282	27933	0.07
mucars	R	ACO	0.00716	27578	27779	0.60
mucars	R	DIME	0.00763	27578	27793	0.40
mucars	R	Expl.-AFA	0.00677	27578	27771	0.00
mucars	R	GDFS	0.00697	27578	27803	0.57
mucars	R	SEFA	0.00660	27578	27764	0.03

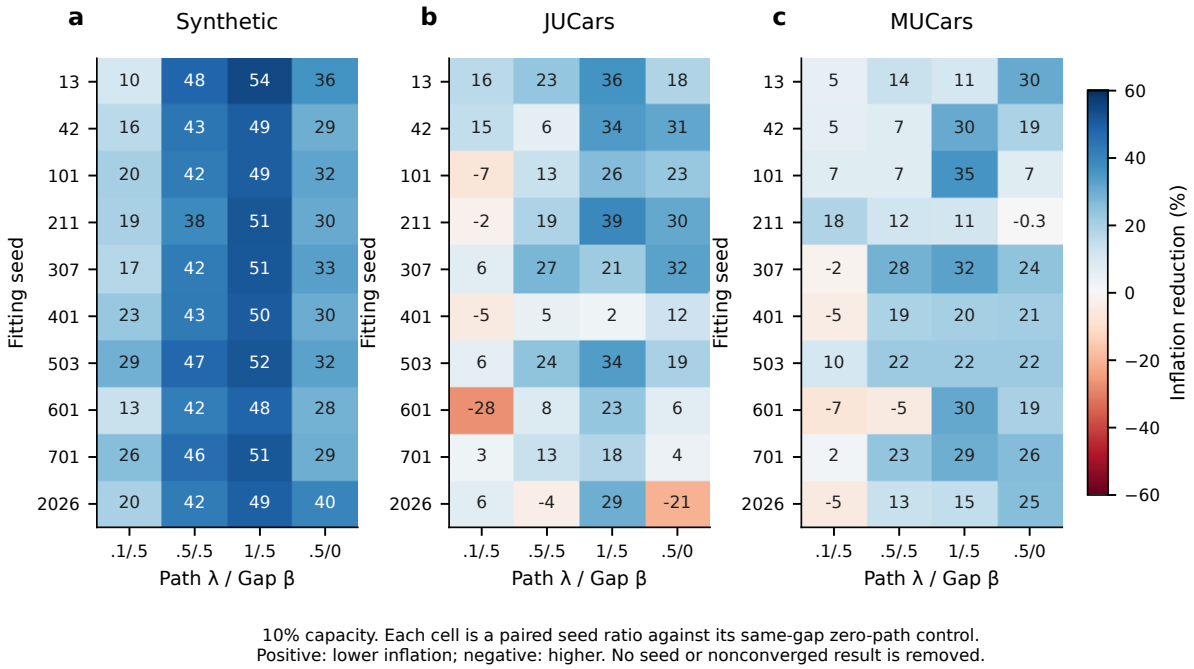


Figure 13: All ten fitting seeds at 10% capacity. Cell values are inflation reductions against each configuration's same-gap zero-path control. Positive values favor RPL; negative values retain adverse seeds. Display values are rounded, retaining one decimal below 1% magnitude; unrounded data are public. No nonconverged outcome is excluded.

I Ten-seed recovery-path diagnostics

Figure 13 retains every fitting seed and all four nonzero path/gap comparisons at primary capacity. Small weights can reverse individual seed effects; their means do not imply per-seed monotonic improvement. All zero-weight controls remain in the public source tables, although their self-difference is identically zero and is not colored as new evidence.

The original three-seed recovery-edge measurement, separate from the ten-seed performance expansion, gave mean positive gain increments of 0.001790 to 0.000693 (synthetic), 0.001648 to 0.001048 (JUCars), and 0.000788 to 0.000450 (MUCars) at $\lambda = 1, \beta = 0.5$. They remained positive, so the zero-edge-penalty premise of Proposition 3 was not achieved. The pilot is not a ten-seed mechanism remeasurement or proof of a unique causal explanation.

The new four-stage ACO audit initially stopped because its independent distance reconstruction subtracted archived float32 bank/query arrays, whereas production promoted both to float64 before distance calculations. The corrected, separately versioned auditor matches that arithmetic contract and passed the full 144-cell chain. The failed receipt, original checker and new source closure remain public; no tolerance, production model or saved result was changed to obtain a pass.

J Detailed historical controls and visual diagnostics

The following displays retain the original feedback and initial-objective comparisons, positive-cost mechanism and capped/uncapped variance diagnostics. They support the corresponding main-text

qualifications and are not additional independent replications. Parameter, cost and historical-control results keep their original sample sizes and uncertainty conventions.

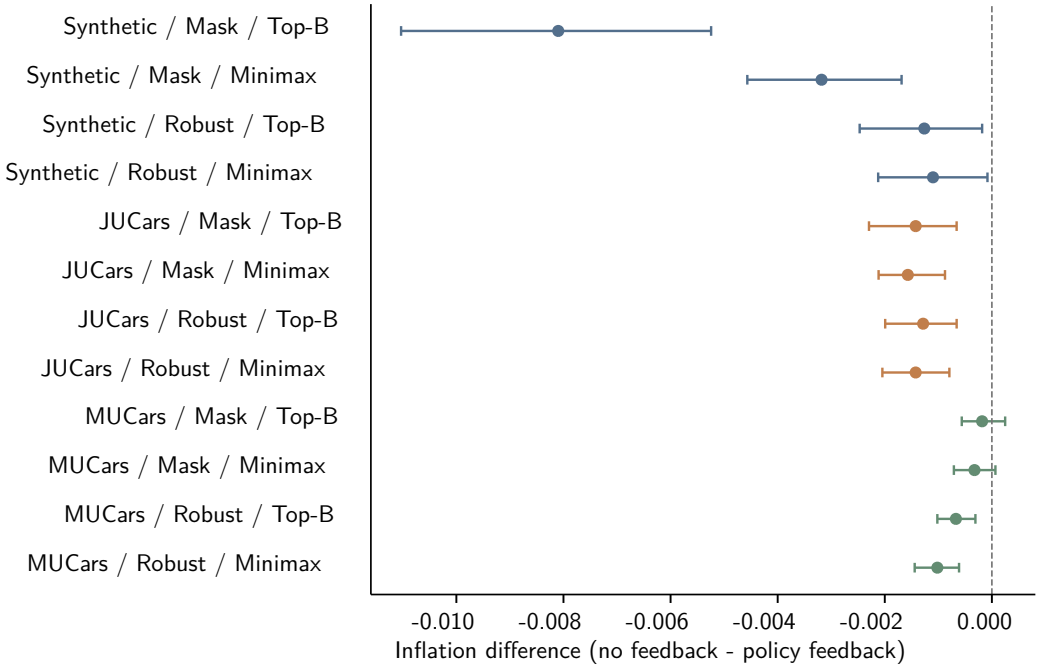


Figure 14: Under the original weak training-response search, policy feedback increases mean inflation in all twelve suppression-acquisition cells at the primary 10% capacity. Points are no-feedback minus policy-feedback paired differences; bars are 95% unadjusted intervals from 1,000 crossed resamples of ten fitting seeds and ten shared batches. Positive differences would favor feedback. All models use the same fixed training records; 1,000 evaluation listings are reused across seeds. The two MUCars mask-objective intervals cross zero. Other scores are retained in the full contrast data.

Family	Comparisons	Critical value	Adverse feedback bands
Feedback inflation	36	3.126	31/36
Feedback, three outcomes	108	3.803	27/36
All primary contrasts/metrics	1,536	4.882	20/36

Table 13: Post-hoc 95% approximate simultaneous sensitivity on the original capped cohorts under the original weak training-response search. These bands do not cover the strengthened-search interventions. The last column counts only the 36 inflation feedback contrasts with an upper endpoint below zero. It is not a confirmatory family-wise guarantee. All point effects and all three family definitions are retained.

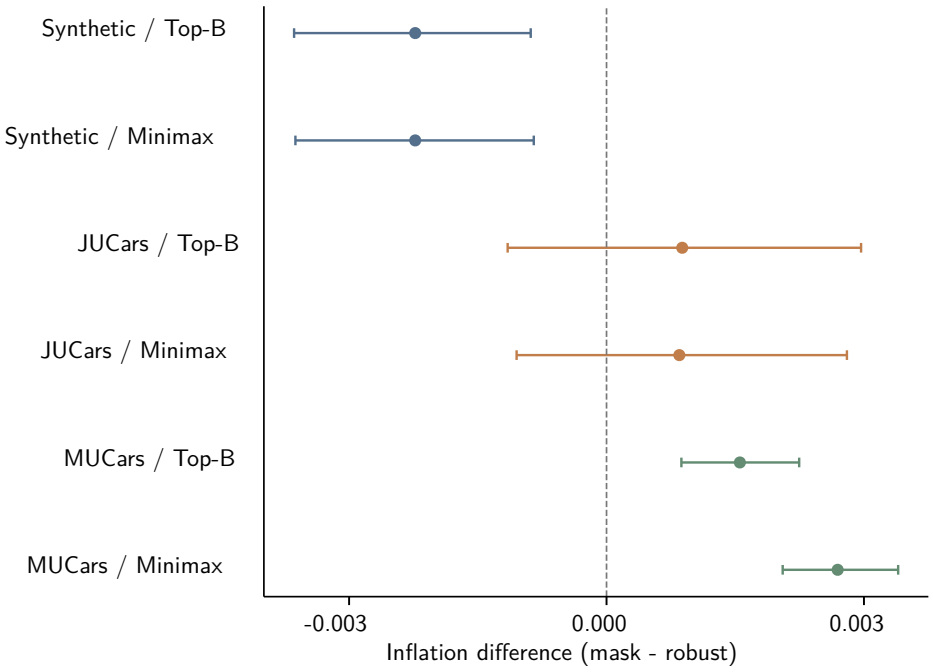


Figure 15: Initial-objective effects depend on the source. With suppression acquisition and no policy feedback, points are mask-objective minus robust-objective inflation differences at 10% capacity. Positive values favor robust initialization. Intervals and repeated-unit definitions match Fig. 14. The source labels and row positions encode cohort identity independently of color. These newly trained objectives are not aliases for the original SAV/static-robust candidates.

Source	Feedback/allocation	Inflation	Honest MAE	Attack MAE	Failure (%)
Synthetic	None/top-B	0.013101	3582.47	3655.51	2.430
Synthetic	None/minimax	0.012868	3582.47	3653.16	1.886
Synthetic	Policy/minimax	0.013967	3589.84	3652.94	2.641
JUCars	None/top-B	0.011112	6909.46	6902.00	6.330
JUCars	None/minimax	0.010432	6909.46	6902.07	5.688
JUCars	Policy/minimax	0.011855	6907.68	6899.03	5.485
MUCars	None/top-B	0.006511	27223.61	27372.61	0.380
MUCars	None/minimax	0.005371	27223.61	27363.32	0.651
MUCars	Policy/minimax	0.006390	27267.93	27441.51	0.542

Table 14: Original weak-search implementation: absolute primary-capacity outcomes for the robust initial objective and suppression acquisition, averaged over ten fitting seeds and the same 1,000 evaluation listings. Inflation is positive expected log-quote inflation; MAEs use the source's own currency, without currency pooling. Failures are per listing, rather than conditional on receiving a query. These are descriptive means; the effect intervals are in the full paired-contrast data.

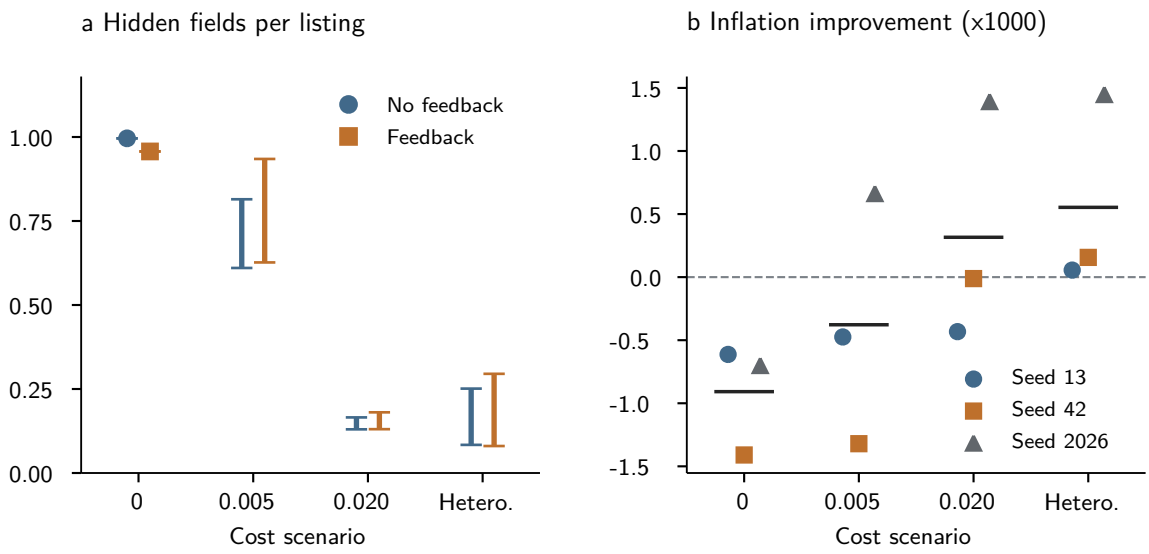


Figure 16: Cost deterrence accompanies the local feedback reversal, whose sign varies across fitting seeds. Synthetic robust/suppression/top-B on the same 1,000 records and three fitting seeds. (a) Zero-cost markers are directly observed mean hidden counts. Positive-cost line segments are utility/regret identification bounds on the mean count, not confidence intervals; no midpoint estimate is asserted. (b) Symbols retain all three paired seed effects, measured as no-feedback minus feedback inflation; horizontal black strokes are seed means and positive values favor feedback. The four categorical scenarios are zero, uniform 0.005, uniform 0.02 and heterogeneous $0.02h_i$, with $h_i \in [0.5, 1.5)$. No interpolation between scenarios or new uncertainty claim is made. All returned nonconverged profiles are retained.

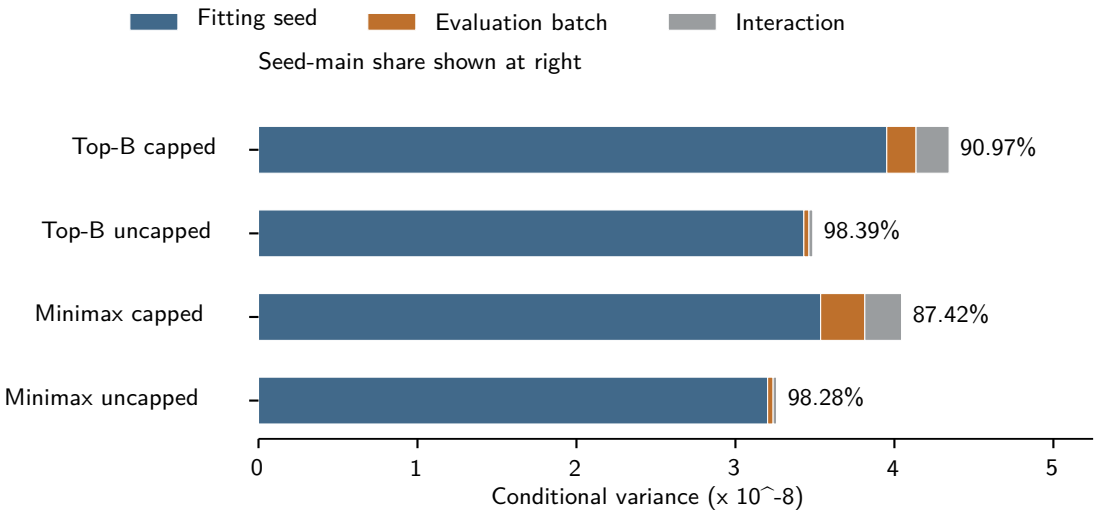


Figure 17: Fitting randomness dominates the remaining conditional uncertainty after full MU-Cars evaluation. Bars show absolute seed-main, batch-main and interaction variances for the masking/suppression feedback contrast under both allocators; labels give the seed-main share. Ten fitting seeds are shared. Capped evaluation uses ten equal 100-record batches; uncapped evaluation uses 107 record-weighted batches containing 10,672 listings. Both the outcome matrix and weights change, so these are distinct conditional designs, not a pure sample-size experiment. Components condition on the fixed training records and fitted models; they do not quantify training-population uncertainty or establish equivalence.

Table 15: Recomputed common-response comparison at 10% capacity with the robust estimator, averaged over three seeds. Positive inflation reduction is favorable; positive attacked-MAE change and failed-query change are unfavorable. Failure change is percentage points per listing, not the conditional failure/query rate.

Source	Rule	Inflation reduction (%)	MAE change (%)	Failure change (pp)
synthetic	suppression	4.82	-0.079	+0.400
synthetic	benefit	7.72	-0.147	+0.467
synthetic	risk	3.70	-0.122	+0.133
synthetic	cdm	1.07	-0.081	+0.160
jucars	suppression	9.70	+0.070	+0.333
jucars	benefit	12.44	+0.080	+0.800
jucars	risk	0.61	+0.002	+0.000
jucars	cdm	7.71	+0.018	+0.300
mucars	suppression	7.62	-0.038	+0.133
mucars	benefit	11.76	-0.030	+0.200
mucars	risk	1.11	+0.005	+0.033
mucars	cdm	4.76	-0.019	+0.067

Cost per hidden field	Inflation improvement	Attack-MAE improvement	Failure improvement (pp)
0	-0.000908	-8.995	-0.533
0.005	-0.000378	+3.043	-0.233
0.02	+0.000316	+14.200	+0.067
Heterogeneous 0.02	+0.000553	+12.198	+0.100

Table 16: Synthetic robust/suppression/top-B feedback sensitivity on the same 1,000 records and three fitting seeds. Improvements are no-feedback minus feedback; positive values mean reductions in inflation, attacked error or failure. Costs are log-quote utility units; MAE uses synthetic price units. All four cost regimes and all individual-seed results are retained; the table is descriptive and post-hoc.

Table 17: Allocation intervention at 10% capacity. Positive inflation reduction favors score-minimax over top-B. Intervals are in log-inflation units, from 1,000 crossed resamples of three fitting seeds and ten shared batches, conditional on fixed training records; they are unadjusted. The final column is minimax minus top-B failed-query probability in percentage points, so a positive value is an adverse change.

Source	Price	Score	Reduction (%)	95% interval	Failure change
jucars	masking	benefit	0.49	$[-0.00005, 0.00027]$	0.204
jucars	masking	risk	-3.26	$[-0.00181, 0.00009]$	0.015
jucars	masking	suppression	-0.30	$[-0.00024, 0.00012]$	-0.604
jucars	robust	benefit	9.99	$[0.00065, 0.00160]$	0.815
jucars	robust	risk	0.52	$[-0.00029, 0.00040]$	-0.011
jucars	robust	suppression	8.00	$[0.00044, 0.00132]$	-0.634
mucars	masking	benefit	-0.51	$[-0.00069, 0.00013]$	-0.043
mucars	masking	risk	-36.94	$[-0.01262, -0.00726]$	-0.000
mucars	masking	suppression	-0.55	$[-0.00066, 0.00009]$	-0.041

Source	Price	Score	Reduction (%)	95% interval	Failure change
mucars	robust	benefit	13.70	$[0.00087, 0.00128]$	0.060
mucars	robust	risk	0.13	$[-0.00024, 0.00024]$	-0.000
mucars	robust	suppression	13.05	$[0.00082, 0.00126]$	0.128
synthetic	masking	benefit	-47.57	$[-0.02905, -0.01810]$	-1.543
synthetic	masking	risk	-49.33	$[-0.02998, -0.01744]$	-1.659
synthetic	masking	suppression	-22.54	$[-0.01618, -0.01067]$	-1.597
synthetic	robust	benefit	-6.65	$[-0.00193, 0.00038]$	-1.186
synthetic	robust	risk	-11.36	$[-0.00221, -0.00037]$	-0.928
synthetic	robust	suppression	2.34	$[-0.00009, 0.00085]$	-0.633

K Additional historical control analyses

The following analyses retain the full original protocols and numerical evidence summarized in the main text. Their original fitting-seed and uncertainty scopes are unchanged.

K.1 Matched factorial effects

Takeaway. Under the original weak training-response search, all 36 primary feedback contrasts increase mean inflation. This documents a failure of that implementation, not a general negative effect of feedback.

Which conclusion is primary? The paper's empirical conclusion is that verification and feedback outcomes depend on the learning and response procedures. The 36/36 count, Table 18, and the controls in Appendix J establish the well-audited original implementation's failure pattern. They are not the headline claim that feedback is intrinsically harmful. The later, four-stage-checked search intervention (Section 7.4) largely removes the synthetic/JUCars mean damage with stronger training search, while MUCars remains adverse in those means. Both findings must be read together; neither identifies weak search as the sole cause nor proves exact-equilibrium benefit. Table 2 separates completed computational checks from independent scientific confirmation.

Inference scope. Every source uses ten fitting seeds and ten shared evaluation batches of 100 listings. The same 1,000 listings are reused across seeds and methods; 4.32 million outcome records are therefore not independent listing observations. We report 1,000 crossed seed–batch bootstrap draws and their 2.5th and 97.5th percentiles. These unadjusted exploratory intervals condition on the fixed training records. They do not cover a new training population or correct for multiple comparisons. The primary capacity supplies 174 factor contrasts and 18 additional joint-policy contrasts, each evaluated on eight metrics. All six capacities and all contrast components remain in the evidence package. Comparisons away from 10% also measure transfer from the fixed training capacity.

Post-hoc simultaneous sensitivity. We supplement the original unadjusted intervals with 20,000 centered crossed bootstrap draws [15]. The family maximum of absolute standardized errors supplies one common critical value; scales are the conditional standard deviations of the crossed bootstrap, not re-estimated inside each draw. Pairing across methods and metrics is retained. We report the 36 inflation feedback contrasts, the same contrasts on three co-primary outcomes (108),

and all 1,536 primary contrast/metric combinations. A source-wise Bonferroni sensitivity avoids requiring cross-source independence. Resampling maxima are established multiple-testing tools [16]; this is a single-step sensitivity analysis, not a transfer of their finite-sample guarantee.

The adverse feedback count falls from 34 unadjusted intervals to 31, 27 and 20 under these three families (Appendix J). Source-wise bands retain 31, 27 and 19. The original 36 adverse point estimates remain unchanged. No family certifies that every contrast is harmful. With only ten seeds and ten batches, approximate simultaneous bands do not establish finite-sample coverage, fresh-cohort confirmation or training-population uncertainty.

Feedback does not explain a beneficial effect in this design. Define a feedback contrast as no-policy-feedback loss minus policy-feedback loss, holding the initial objective, score and allocator fixed. For positive expected log-quote inflation, all 36 point estimates at 10% were negative; 34 unadjusted intervals excluded zero in the unfavorable direction. The two intervals crossing zero corresponded to MUCars with the mask initial objective and suppression acquisition. No feedback contrast passed the point gate of at least 10% inflation reduction with at most 5% honest-MAE increase. This is a negative result for the implemented two-round, epoch-refreshed, weak-response-search training procedure. The strong-search means in Section 7.4 narrow its interpretation; they do not alter the original counts.

Table 18 summarizes all primary directions.

Source	Feedback harms	Harm intervals	Robust-init gains	Joint gates
Synthetic	12/12	12/12	4/12	0/6
JUCars	12/12	12/12	12/12	0/6
MUCars	12/12	10/12	12/12	0/6

Table 18: Original weak-search implementation: primary-capacity factorial diagnostics. These counts do not apply to strong-search retraining (Section 7.4). Counts refer to positive expected log-quote inflation. Feedback harms and robust-initialization gains are point-estimate directions. Harm intervals are 95% unadjusted intervals excluding zero adversely for the capped primary evaluation. The uncapped MUCars sensitivity also excludes zero in 10/12 intervals (Section 7.7). A joint gate compares no feedback/top-B with policy feedback/minimax under a fixed initial objective and score, requiring at least 10% inflation reduction and at most 5% honest-MAE increase. Counts are descriptive across correlated contrasts, not independent trials.

Appendix J displays the suppression-acquisition comparisons.

Initial-objective effects depend on the source. The robust initial objective lowered mean inflation in all twelve MUCars conditional contrasts. Reductions ranged from 12.43% to 33.40%, all twelve unadjusted intervals excluded zero favorably, and all twelve passed the point gate. Honest MAE also decreased in these comparisons. On JUCars, twelve point estimates favored the robust objective but only two intervals excluded zero favorably. On synthetic data, only four point estimates favored it, and six intervals excluded zero in the adverse direction. Thus the controlled initial-objective factor has a stronger source-dependent signature than a beneficial feedback effect. These are newly trained matched models, so they do not retrospectively identify the cause of the original SAV/static-robust comparison.

Appendix J shows the no-feedback suppression comparisons.

A positive interaction can mean less harm. For synthetic data with the mask objective and suppression acquisition, the feedback–allocation interaction was $D = 0.004917$, with an unadjusted interval $[0.002378, 0.007586]$. Feedback increased inflation under both allocators: from 0.010871 to 0.018969 under top-B, and from 0.010638 to 0.013818 under minimax. The positive interaction measures the smaller increase under minimax; it does not show that minimax feedback improved on its no-feedback control. Across all sources, none of eighteen joint changes from no feedback/top-B to policy feedback/minimax passed the point gate. The three favorable MUCars joint point estimates were small and their intervals crossed zero. Inflation, honest error, attacked error and failed-query effects must therefore be read jointly rather than inferred from the interaction sign.

Mechanism and practical boundary. The factorial experiment isolates implemented training and allocation choices, but does not identify a unique cause of its feedback damage. Section 7.4 intervenes on response-search strength and score-refresh frequency; stronger search removes some damage, while capacity transfer is not separately identified. These computationally checked development interventions qualify the negative result without establishing a sole cause. The appropriate practical comparison is the matched no-feedback control with the same initial objective, score and capacity. Choosing feedback from a positive interaction alone is unsupported. A risk score is also not a universal fallback: its feedback-trained cells are among the adverse cases. Performance on a new cohort requires renewed multiobjective validation; point Pareto flags in the present cohort are descriptive, not probabilistic dominance guarantees.

Absolute outcomes qualify an inflation-only ranking. Appendix J holds the robust initial objective and suppression acquisition fixed. Minimax feedback increased JUCars inflation from 0.010432 to 0.011855 while reducing attacked MAE from 6902.07 to 6899.03 and lowering query failures. On MUCars, the same feedback contrast increased both inflation and attacked error. Thus improving prediction error need not reduce the seller's gain over the model's own honest quote. Of the 432 source/capacity/cell settings after aggregating seeds, 37 contained a nonconverged batch; eight of these were synthetic primary-capacity settings. The maximum recorded remaining regret was 0.177513 log units. All were retained.

K.2 Strong controls and the original candidate comparison

The original SAV study supplies a comparator check rather than a factorial attribution. Its ten-seed extension covers 30 source–seed pairs, 1,860 cells and 1.86 million repeated outcomes. At 10% capacity, SAV inflation was 0.021175 on synthetic data, 0.014988 on JUCars and 0.007340 on MUCars; the corresponding static-robust values were 0.012084, 0.012082 and 0.006511. The static-robust-minus-SAV crossed intervals were $[-0.012355, -0.006323]$, $[-0.007025, 0.000777]$ and $[-0.001870, 0.000210]$, respectively. These comparisons do not support consistent SAV superiority; intervals including zero also do not establish equivalence. Appendix C retains all five original comparators for each source, including strong benefit-head controls and weaker masking/no-gap variants. These original models must not be substituted for the matched factorial objectives.

Competitive controls materially change the original-method story. In the audited three-seed ordinary-benefit extension, reference-minus-SAV inflation differences are -0.00998 on synthetic and -0.00432 on JUCars; their conditional intervals are $[-0.01260, -0.00791]$ and $[-0.00598, -0.00285]$. The ACO-1 control also beats SAV on these sources, with SAV-minus-ACO differences 0.00995 and 0.00431 and intervals $[0.00725, 0.01345]$ and $[0.00291, 0.00588]$. These are separate original-protocol three-seed comparisons, not new ten-seed matched-factorial effects. A universal SAV advantage is not reproduced against these strong controls. MUCars remains conditional: ordinary benefit favors

SAV by 0.00089, while the ACO comparison spans zero; the largest-cap equal-tuning experiment favors SAV on MUCars but reverses on the other sources. Appendix E preserves all fitting, tuning and adaptation boundaries. The main implication is to compare strategy-aware valuation with competitive predictive acquisition before attributing a gain to strategic feedback.

K.3 Allocation changes on frozen models

Takeaway. Score-minimax allocation can help with a robust price model, but the same intervention can worsen inflation with a masking model. It is not a universal replacement for top-B.

This separate experiment retains the original three seeds and frozen price models. It crosses three sources, two price models, three public acquisition scores and two allocators at 10% capacity: 108 cells and 108,000 outcome rows. Seller responses are recomputed for each allocator. The separate evaluator reconstructs 1,080 method/batch cells, 1,134,324 legal unilateral deviations, model outputs, allocations and expected-error arithmetic. A second checker rebuilds all eighteen contrasts, five metrics, resampling intervals, decision gates and pointwise dominance flags from the raw records. This is an allocation intervention conditional on frozen models, not the full feedback-training study.

Appendix J retains all eighteen contrasts. Only robust-benefit and robust-suppression on MUCars meet the inherited heuristic point criteria of at least 10% inflation reduction and at most 5% honest-MAE increase. They reduce inflation by 13.70% and 13.05%, but increase failed-query probability by 0.060 and 0.128 percentage points. Honest MAE is unchanged in every contrast under these replay semantics. Ten of eighteen contrasts increase inflation; four have a positive lower endpoint for the inflation reduction. JUCars robust-benefit improves by 9.985%, which does not meet the 10% threshold. No threshold is relaxed after observing this near miss.

The multiobjective results qualify the apparent gains. MUCars robust-benefit and robust-suppression also reduce expected attacked MAE by 7.17 and 7.67 source-currency units, but their failed-query increases prevent Pareto dominance. JUCars robust-benefit and robust-suppression increase attacked MAE by 2.06 and 0.62 units despite reducing inflation. Only JUCars robust-risk and synthetic robust-suppression weakly dominate top-B on all four point metrics; neither has an inflation interval excluding zero. These point comparisons use a numerical tie tolerance, not a claim of practical equivalence.

Ten top-B response batches fail the convergence criterion (six synthetic masking-benefit and four synthetic masking-suppression); their maximum unilateral regret is 0.095752. The minimax responses have zero recorded final regret over the enumerated legal unilateral actions. All batches are retained. Thus the intervention includes its response-search consequences, and the observed reversals cannot be attributed solely to equilibrium allocation quality. The score objective remains a surrogate and does not prove general incentive optimality. The completed factorial study evaluates whether training feedback changes these conditional effects.

K.4 Continuous effects and decision-threshold sensitivity

The inherited 10% inflation-reduction/5% honest-MAE gate is a heuristic, not an externally validated minimum important difference or a preregistration claim. We retain it to avoid retrospective threshold selection, report the continuous effects, and vary inflation cutoffs over 0, 5, 10, 15 and 20% and honest-error allowances over 0, 1, 5 and 10%.

At a 5% honest-error allowance, the frozen allocation study has 8/18, 4/18, 2/18, 0/18 and 0/18 passing contrasts across the respective inflation cutoffs. Its JUCars robust-benefit reduction is approximately 9.985%, so classifying it below a 10% gate is not a substantive discontinuity in

usefulness. In the matched joint-policy family, the zero-cutoff gate retains 3/18 contrasts, all small MUCars gains; 0/18 pass any inflation cutoff of 5% or more. Full grids and unrounded effects are supplied. A deployment threshold requires a platform's error/query-cost preferences, which this retrospective study has not elicited.

K.5 Positive costs change the conditional feedback ranking

We recompute responses under per-hidden-field costs 0.005 and 0.02 in log-quote utility units, and heterogeneous costs $0.02h_i$, where a fixed listing hash gives $h_i \in [0.5, 1.5)$. These are stipulated post-hoc sensitivities, not measured human effort or monetary sanctions. All 108 positive-cost cells retain the robust initial objective and suppression score, three original fitting seeds, both feedback settings and both allocators. Four response starts and twenty sweeps are unchanged. Costs enter report choice; they are not simply subtracted from saved zero-cost outcomes. Independent algebraic replay and every legal unilateral utility corroborate the returned profiles.

Appendix J shows a ranking reversal on synthetic top-B. Uniform 0.02 and heterogeneous costs have lower mean inflation with feedback than without it; the heterogeneous direction is positive for all three seeds, while the uniform 0.02 direction is positive for only one. At heterogeneous cost, feedback also lowers honest MAE by 4.293 synthetic price units, attacked MAE by 12.198 units and failed queries by 0.100 percentage points. This is descriptive mean Pareto improvement within that declared subset, not an uncertainty-qualified or universal dominance result. Failure remains high in absolute terms (9.4% without and 9.3% with feedback per listing).

Appendix J separates concealment deterrence from the paired feedback effects.

The reversal accompanies strong concealment deterrence. At uniform cost 0.02, synthetic top-B mean hidden-field counts are bounded by $[0.130, 0.165]$ without feedback and $[0.130, 0.181]$ with it, versus directly observed zero-cost means 0.996 and 0.957. These bounds retain nonconverged profiles rather than treating the inflation–net-deviation gap as exact cost. The legal net-deviation envelope falls by 77.5% and 81.1%, respectively. Failed queries reach 9.47% and 9.40% per listing; at 10% capacity, at least 94.7% and 94.0% of actual queries fail. Thus the local ranking reversal coexists with suppressed concealment and predominantly unsuccessful verification; it does not isolate improved feedback quality as its cause. Constructively, deterrence moves this subset toward ordinary valuation: at both 0.02 cost specifications, feedback no longer worsens mean inflation and lowers mean honest error, attacked error and query failure. This is a conditional descriptive benefit, not a fully nonstrategic limit: residual concealment remains positive. Appendix G gives the utility-based bounds.

JUCars and MUCars retain adverse mean inflation directions for both allocators under every declared cost regime in this subset. Twenty positive-cost seed/cell batches remain nonconverged and are retained. Therefore the original 36/36 zero-cost direction describes its evaluated regime, not a cost-invariant property of feedback training. The full three-source cost table is in the evidence package; no winning cost distribution is substituted for the zero-cost primary experiment.

K.6 Full-validation evaluation coverage

An additional frozen-model evaluation removes both real-source caps: all 1,374 JUCars and 10,672 MUCars validation records form fourteen and 107 batches, respectively, using the same ten fitting seeds. Final batches contain 74 and 72 records, each with capacity seven. All 480 cells, 29,040 seed/cell batches and 2,891,040 repeated outcomes pass algebraic allocation, bit-transition, metric and complete unilateral-utility checks, covering 31,269,840 legal deviations. All returned real-source batches meet the finite unilateral-regret tolerance; this does not certify arbitrary actions or globally

optimal joint responses. The first ten batches reuse the original prediction/outcome bytes through symlinks, with original receipt provenance retained in the extended bindings; only the remaining batches are newly evaluated. No model is retrained.

All 24 real-source feedback inflation point estimates remain adverse. Ten-thousand-draw exploratory intervals exclude zero adversely for all twelve JUCars comparisons and ten MUCars comparisons; both MUCars masking-objective, suppression-score comparisons span zero. These unadjusted intervals weight batches by record counts, including the partial batches. They do not add independent training records or make either retrospective source confirmatory. The original capped factorial remains the declared primary comparison.

An exact conditional seed/batch variance decomposition, including weighted partial batches, attributes 98.39% and 98.28% to fitting-seed main effects for the uncapped MUCars masking/suppression top-B and minimax contrasts. Their capped shares are 90.97% and 87.42%; these compare different cohort and weighting designs, not a pure replication-size change. Interval-width ratios alone do not identify the components. Under the empirical conditional design, thirty instead of ten independent fitting seeds would project standard deviations to approximately 58% of their current values, while eliminating only batch resampling noise leaves approximately 99%. These are design projections, not newly trained results. The stable small estimates and crossing-zero intervals establish neither zero effect nor equivalence; fitting randomness on fixed training records also does not quantify training-population uncertainty. Details are in Appendix G.

Appendix J shows absolute components alongside their shares.

The accompanying `audit-protocol/` artifact provides a machine-readable worked contract, an execution protocol and a standard-library validator of six directly bound evidence files. Six invalid-contract controls are rejected. It is an explicit post-hoc protocol, not preregistration or a certificate of scientific validity or independent end-to-end reproduction.

K.7 Recomputed cap responses and multiobjective trade-offs

Takeaway. The cap's inflation reductions survive the saved response heuristic in this study, but attacked error and failed-query rates can worsen. This is an empirical result, not an extension of the fixed-peer theorem.

We recomputed reports for every source/seed/estimator/capacity/rule cell, using the original four starts and up to 20 sweeps, frozen public scores and unchanged mixture weights. The uncapped baseline uses the separately corroborated original responses. All 432 new cells and 432,000 listing outcomes were replayed by an separate full-sort auditor. It checked 5,300 supported complete-rule batch allocations and 5,570,112 legal rule–deviation outcomes. The auditor reconstructs final quotations, currency absolute errors, query failures and remaining unilateral regret; it does not certify a globally optimal joint response or every search step.

Appendix J reports the primary robust-estimator comparison.

Figure 8 displays the two principal trade-off coordinates.

The consequential finding is the joint cost: attacked MAE increases in 94 of 432 recomputed-response cells and failed-query probability increases in 145, while honest MAE stays unchanged. Inflation falls in 279 cells and ties in 153; attacked MAE falls in 185 and ties in 153, and query failures fall in 4 and tie in 283. All counts remain descriptive. The fixed-peer theorem already establishes pointwise cap monotonicity and is not an empirical effectiveness test. Here peers' responses are recomputed, so cross-profile monotonicity is not supplied by that theorem either. The useful question is whether any inflation reduction survives the attacked-error and query-failure trade-offs, not how many cells exhibit the cap's intended quote reduction.

Thirty-three capped cells contain at least one batch with remaining unilateral regret above the

convergence tolerance, compared with 34 original cells. The maximum capped regret is 0.204423 log units. We retain these cases. In the primary robust benefit comparison, one synthetic seed has one nonconverged batch, with maximum regret 0.029232; the JUCars and MUCars primary benefit batches pass the specified local regret check. Therefore no result is described as a certified market equilibrium.

For the 12 primary source/rule contrasts, we compute unadjusted 95% percentile intervals from 1,000 paired resamples of the ten shared batches, first averaging the three fixed fitted-seed outcomes within each batch. These are conditional evaluation-batch intervals, not training-population intervals or 30 independent batch replicates. For benefit, the absolute inflation-reduction intervals are $[0.000388, 0.001252]$ (synthetic), $[0.001207, 0.001586]$ (JUCars), and $[0.000842, 0.000995]$ (MUCars). The JUCars and MUCars point reductions satisfy the original 10% inflation/5% honest-error criterion with a positive conditional interval, but that criterion does not constrain attacked error or query failures. In particular, JUCars attacked MAE increases by 0.080%.

Risk had no change in the earlier primary fixed-profile table, but can change under the newly computed profiles. The comparisons condition on different peer reports, so these findings are compatible. The existing fixed-profile theorem must not be used as a common-response or Pareto-dominance guarantee.

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